

SPATIO TEMPORAL VARIABILITY OF URBAN HEAT ISLAND THROUGH REMOTE SENSING AND MACHINE LEARNING TO ASSESS SOCIO-ECOLOGICAL EFFECTS ON SUSTAINABLE DEVELOPMENT



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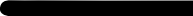
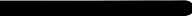
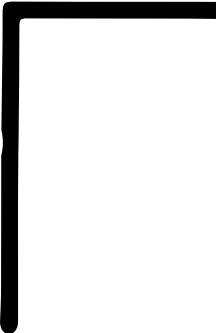
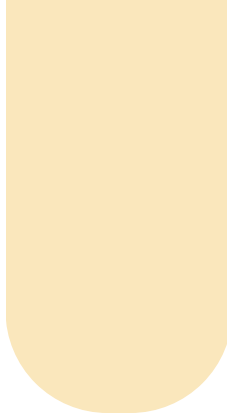
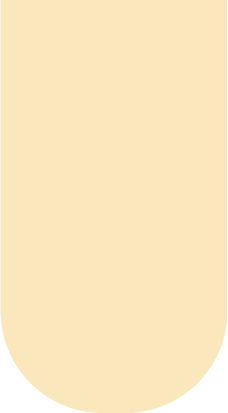
DR . S . Adishkumar ,
Head of the department ,
Department of Civil Engineering ,

TEAM MEMBERS:

Manoselvan R
Mohammed Bazil SM
Mubeena Miyammal M
Mohamed Abdul Latheef M

INTRODUCTION

- Thoothukudi is a rapidly urbanizing coastal city, experiencing significant land use and land cover transformation.
- The presence of industrial zones, port activities, and thermal power plants contributes to elevated surface temperatures.
- Sparse vegetation and expanding built-up areas intensify Urban Heat Island effects.
- Coastal climate variability makes Thoothukudi ideal for studying spatio temporal Urban Heat Island dynamics.
- Urban heat stress poses social and ecological risks, affecting human comfort, coastal ecosystems, and urban sustainability.



LITERATURE REVIEW

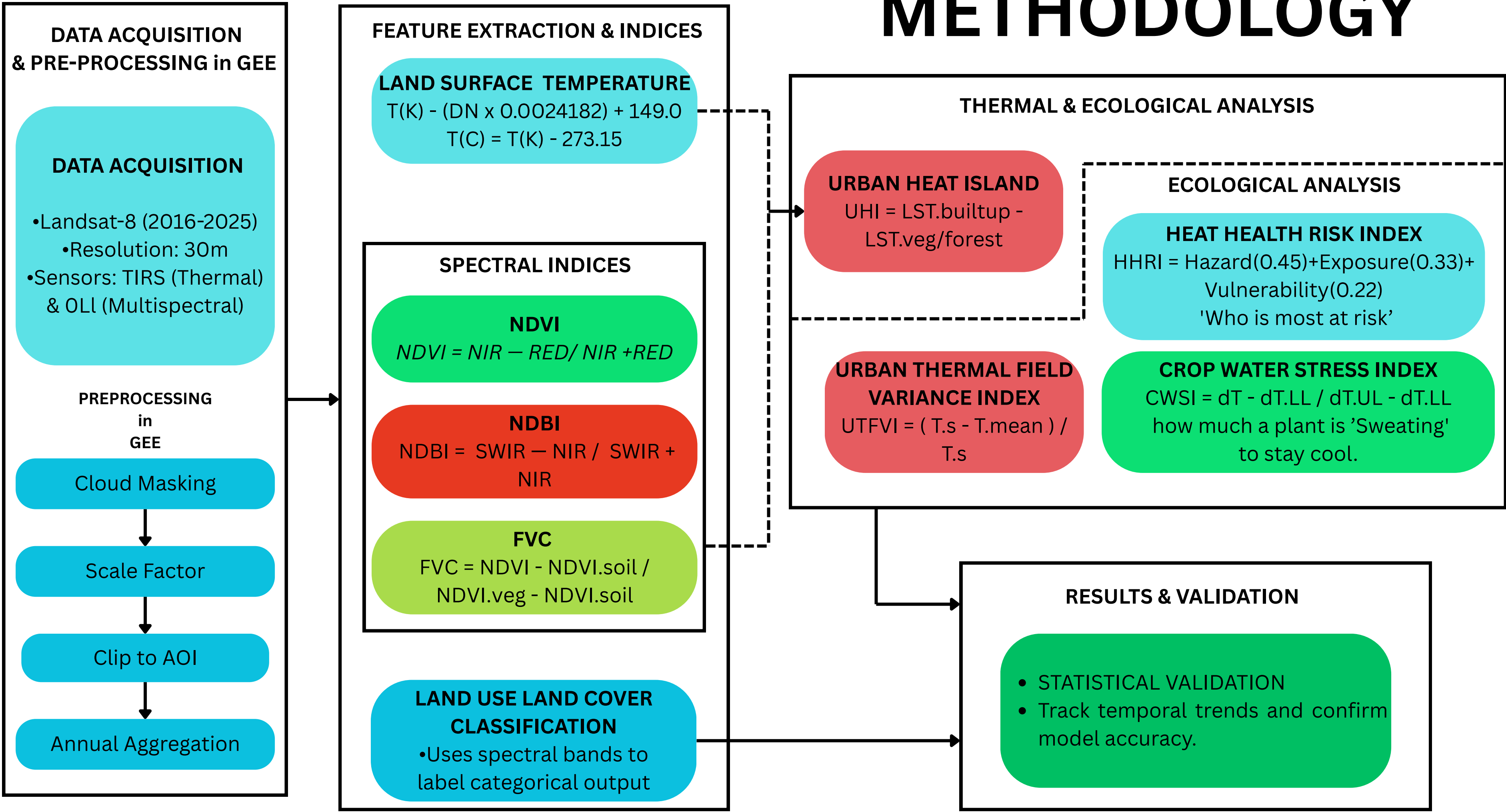


TITLE	STUDY AREA	DATA SOURCE	METHODOLOGY	TECHNIQUE	AUTHOR	INFERENCE
Assessing urban heat island effects in Addis Ababa, Ethiopia using remote sensing and LULC analysis: Implications for sustainable development	Addis Ababa, Ethiopia	Landsat-8 (1990 - 2023)	LST, LULC, NDVI, NDBI, FVC.	Rgression Model (OLS,GWR)	Tsegaye Ayele Tadasse (2025)	This is a sophisticated statistical study. By using Geographically Weighted Regression (GWR), the researchers aren't just saying "it's hot"; they are quantifying exactly how much a 1% increase in building density (NDBI) increases the temperature compared to a 1% increase in vegetation (NDVI).
A study of urban heat island effects using remote sensing and GIS techniques in Kancheepuram, Tamil Nadu, India	Kancheepuram, India	Landsat-4,5,8 (1998, 2010, 2020)	LULC, LST, NDVI, HOTSPOT ANALYSIS	-	Rahman Khatibi (2023)	Using a long-term range (1998–2020), this study likely demonstrates the "Hotspot Creep"—how heat islands expand from a small city center to the outskirts as the town grows into a major industrial/temple hub.
Geospatial evaluation of normalized difference vegetation index (NDVI) and urban heat island: a spatio-temporal study of Gandhinagar City, Gujarat, India	Gandhinagar, India	Modis (2008, 2013)	LST, NDVI, IMPACT MAP	Positive and Negative Correlation	Timsi Modi (2024)	This study highlights the temporal impact of urban expansion over 15 years. By comparing 2003 and 2018, the research likely identifies how the conversion of planned green spaces into built-up areas directly correlates with a rising LST in Gujarat’s capital

Exploring the land-use urban heat island nexus under climate change conditions using machine learning approach: A spatio-temporal analysis of remotely sensed data	Emilia, Italy	Landsat-5,8 (1991 - 2021)	LST, UTFVI, YEARLY TREND ANALYSIS	Gradient Boosting Modelling	Priyanka Rao (2023)	This represents the "modern" approach. By using Gradient Boosting (GB) modeling, the study suggests that UHI patterns are predictable. It implies that we can use historical Landsat data to train AI to forecast future "Heat Waves" with high accuracy
Modelling urban heat island using remote sensing indices in Tiruppur municipal corporation	Tiruppur, India	Landsat-5,8 (1991,2001,2011,2021)	LST, LULC, NDVI, NDBI, UTFVI.	Support Vector Machine	Masilamani Palanisamy (2023)	By focusing on the Urban Thermal Field Variance Index (UTFVI), this paper evaluates the "quality" of the urban environment. It likely concludes that the industrial nature of Tirupur creates intense thermal hotspots that require specific "cool roof" or "green belt" interventions.
Evaluating heat health risk in Indian cities: Geospatial and socio-ecological analysis	Major cities in India	Landsat-8 (2020)	LST, NDVI, SOLAR RADIATION, HUMIDITY, HHRI	Relative Weights	Kaushik Mandvikar (2024)	This paper moves beyond physics and into public health. By incorporating humidity and wind speed, it infers that UHI is not just a climate issue but a "Health Risk." It likely identifies which urban populations are most vulnerable to heat-related illnesses based on their local environment.
Spatio-temporal Assessment of Land Surface Temperature, Vegetation and Crop Water Stress for Sustainable Land Management in Upper Brahmaputra Valley, Assam	Upper Brahmaputra Valley, Assam, India	Modis Terra 6.1 (2001, 2021)	LST, NDVI, HOTSPOT ANALYSIS, CWSI	Pearson's Correlation coefficient	Arunima Nandi (2025)	This paper bridges the gap between UHI and Agriculture. By using the Crop Water Stress Index (CWSI), it infers that rising surface temperatures are actively dehydrating the vegetation in Assam, potentially leading to lower crop yields in the valley.

Interpreting complex relationships between urban and meteorological factors and street-level urban heat islands: Application of random forest and SHAP method	Seoul, Korea	Modis , Landsat8 (2013 - 2022)	LST, UHI STREET LEVEL ANALYSIS	Random Forest	Tageui Hong (2025)	By pairing Random Forest with explainable AI, this paper identifies the specific urban "DNA" causing local heat islands. It transforms broad satellite imagery into a precise, actionable toolkit for cooling down our streets.
Comparing daily and 8-day MODIS land surface temperature data for urban heat island assessment using random forest modeling in data-limited regions	France	Modis (2014 - 2023)	LULC, LST, NDVI, NDBI, NDWI	Random Forest	Vannesah (2026)	This research proves that 8-day MODIS cycles are a powerhouse for tracking urban heat when paired with the right environmental layers.
Assessment of Urban Heat Island using GIS and Remote Sensing – A Case Study of Chennai City, India	Chennai	Landsat 7,8 (2000, 2010, 2020)	LST,NDVI,NDBI, LULC,LSE	Relationship Between LST and INDICES	Mohamed Faizan (2020)	By swapping greenery for concrete, Chennai's rapid growth has created "heat islands" where urban temperatures soar. The study proves that satellite data is a vital tool for planning the greener, cooler cities we need to survive.

METHODOLOGY



DATA COLLECTION - LANDSAT 8

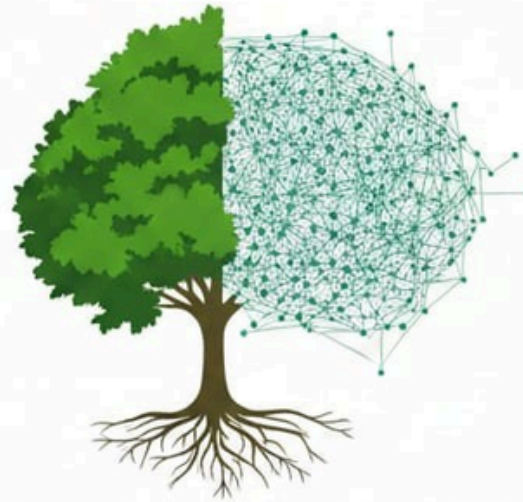
- Total Synergy: Harmonized OLI & TIRS for landscape + heat.
- High Sensitivity: 12-bit depth for maximum radiometric detail.
- Sharp Resolution: Precise 30m/100m spatial grain.
- UHI Insights: 16-day cycles to capture long-term urban trends.
- Instant Indices: Single-workflow NDVI, NDBI, and LST.

OBJECTIVES

1. To generate Landsat8 TIRS and OLI sensors annual cloudfree composites for the study area via Google Earth Engine.
2. To quantify thermal indices namely LST and biophysical indices namely NDVI, NDBI, FVC to track landcover changes.
3. To categorize the ecological quality of the urban environment based on heat island dynamics namely Urban Heat Island (UHI) and the Urban Thermal Field Variance Index (UTFVI) by random forest technique.
4. To Model Ecological Risk through Heat Health Risk Index (HHRI) and Crop Water Stress Index (CWSI).
5. To Perform statistical validation of the derived models and suggest datadriven policies and urban resilience.

THOOTHUKUDI: MITIGATING URBAN HEAT

- Rapid industrial growth, port activities, and urban expansion intensify Urban Heat Island effect
- Hot coastal climate + reduced green cover = increased thermal stress
- Traditional measurements miss detailed heat variations
- Remote sensing & AI enable accurate heat detection



- Rising urban heat threatens public health zones for climate-resilient planning



FIG : 1 



FIG : 2

Urban Heat Island Effect: Concepts & Solutions

🌡️ Urban Heat Island (UHI)

Urban areas exhibit higher temperatures than surrounding rural regions built-up surfaces and reduced vegetation.



🛰️ Remote Sensing

Satellite data is used to extract land surface temperature, and spectral indices for heat analysis.



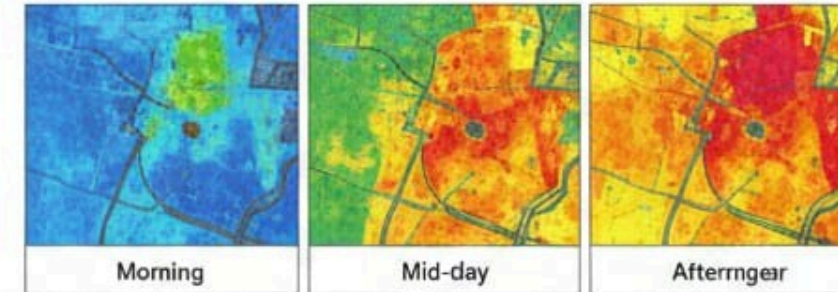
🌿 Social & Ecological Impacts

UHI affects human thermal comfort, public health, vegetation stress, and urban ecosystems.



🌐 Spatio-Temporal Variability

Urban heat intensity varies across locations and changes over time due to land use and climatic factors.



🏠 Machine Learning

ML algorithms identify patterns, classify heat zones, and predict future UHI hotspots efficiently.



🌱 Sustainable Development

Understanding UHI supports climate-resilient planning, infrastructure, and sustainable urban growth.



FIG: 4

SCOPE

The project focuses on analyzing the Spatio-temporal variability of Urban Heat Islands using advanced remote sensing techniques. The scope is divided into four key phases

- Leveraging Google Earth Engine for efficient acquisition and preprocessing of multi-temporal satellite datasets.
- Extracting spatial features and calculating environmental indices to characterize urban landscapes.
- Performing a deep dive into Thermal and Ecological Analysis to understand the relationship between urban growth and heat patterns.
- Ensuring accuracy through rigorous Result Verification and validation against ground-truth or secondary data sources.

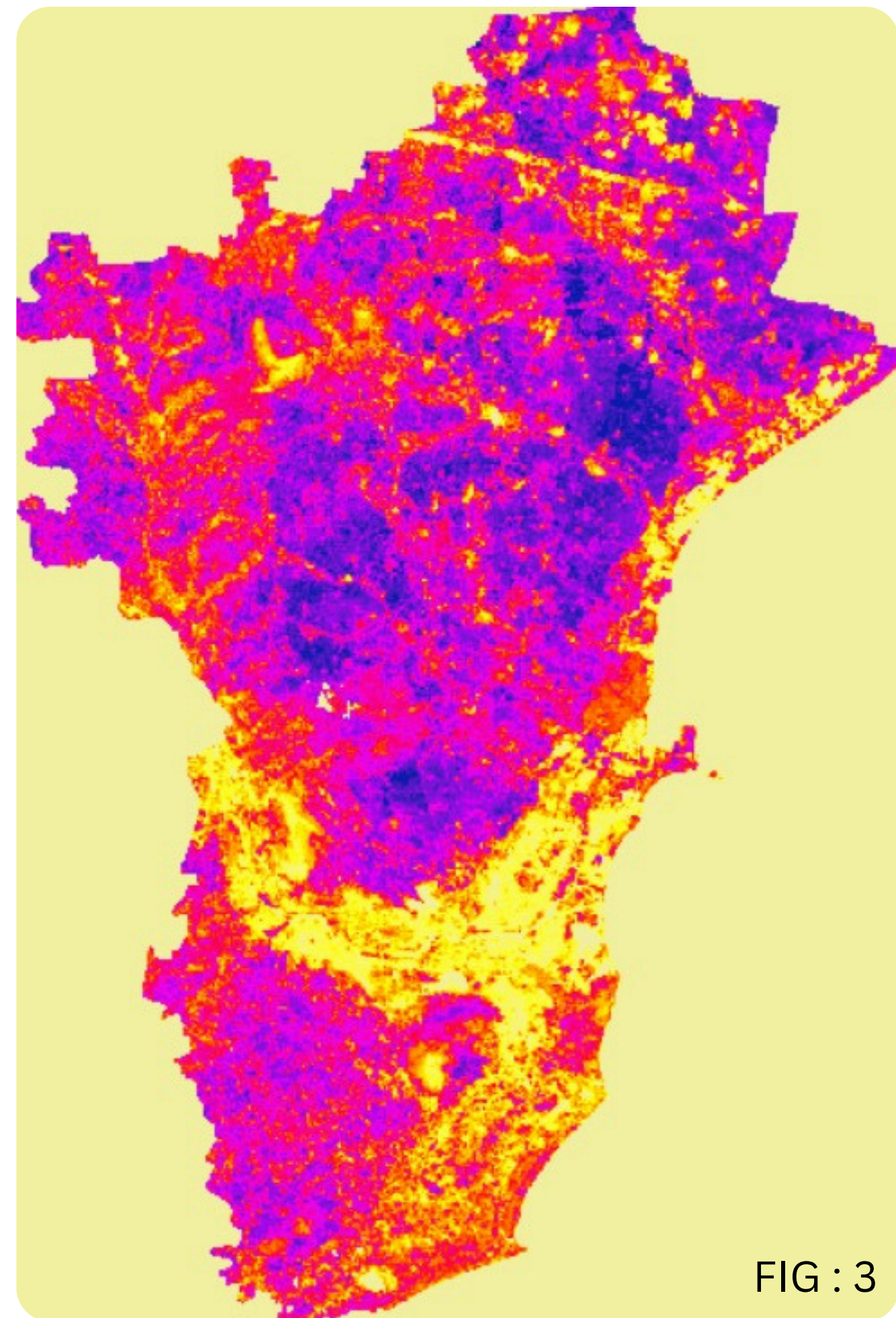
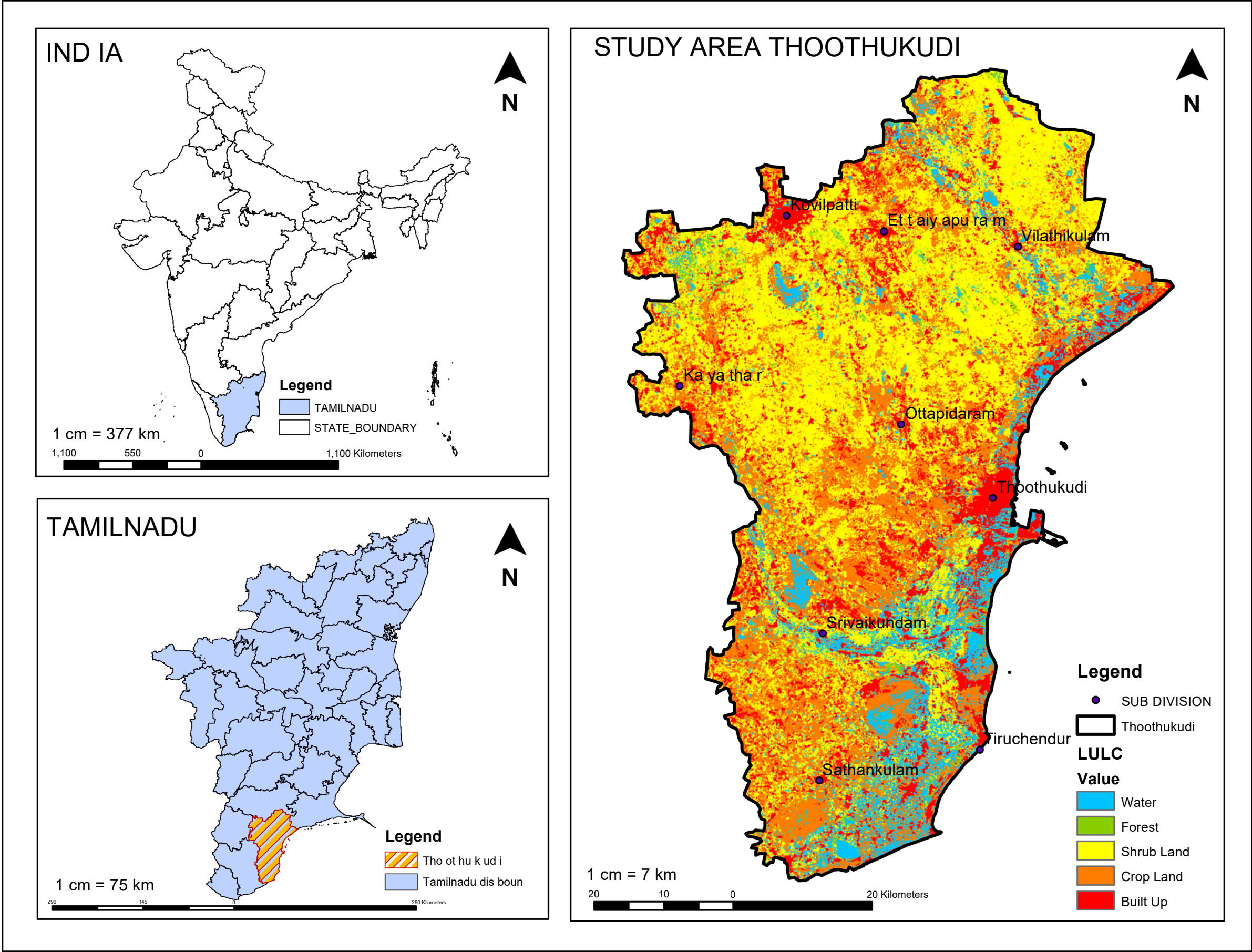


FIG : 3

STUDY AREA - THOOTHUKUDI



PROBLEM IDENTIFIED

The Industrial "Furnace" Effect

Problem: Massive port activities and thermal power plants create 24/7 heat traps.

Impact: Industrial zones act as "core engines" for the Urban Heat Island.

The Coastal Humidity "Multiplier"

Problem: High humidity + UHI = Dangerous "Real Feel" temperatures.

Impact: Coastal breezes are no longer enough to cool the growing concrete landscape.

Greenery vs. Grayness

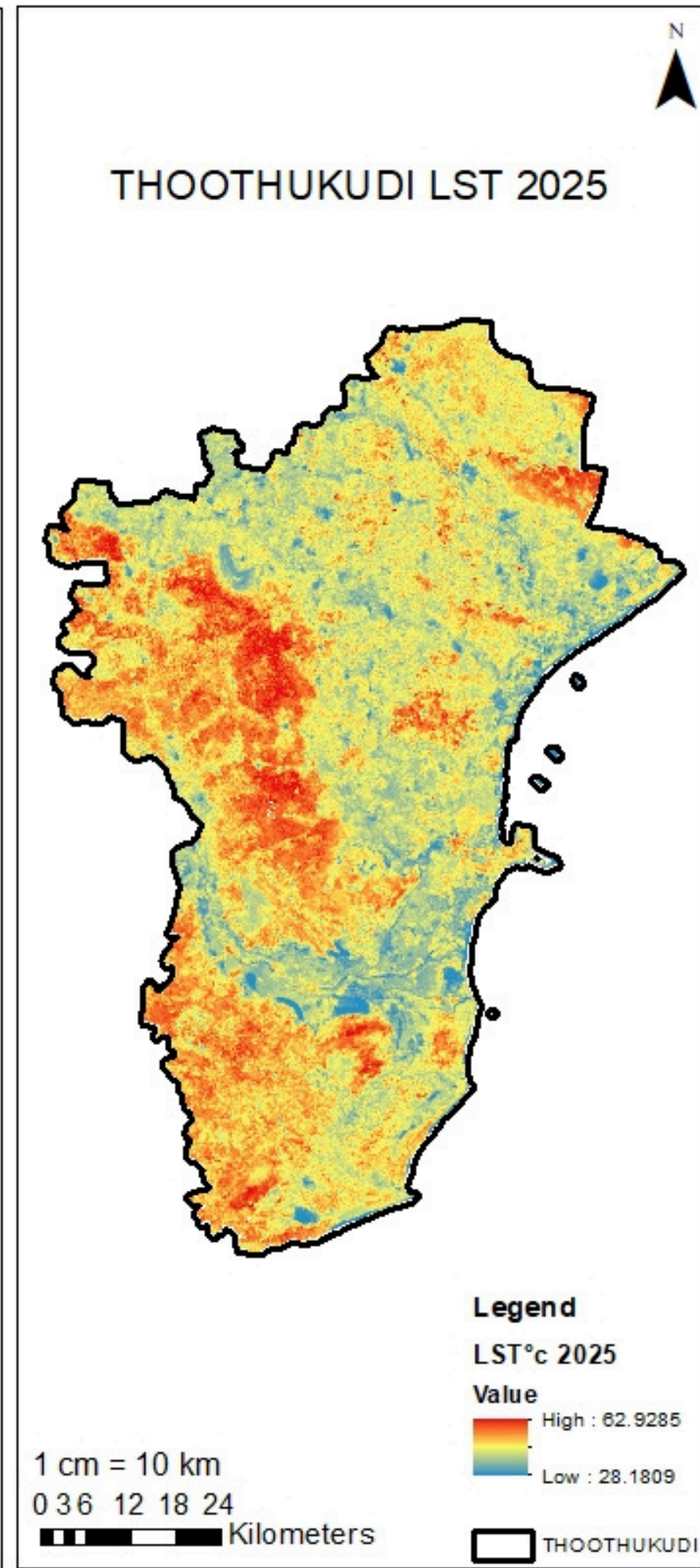
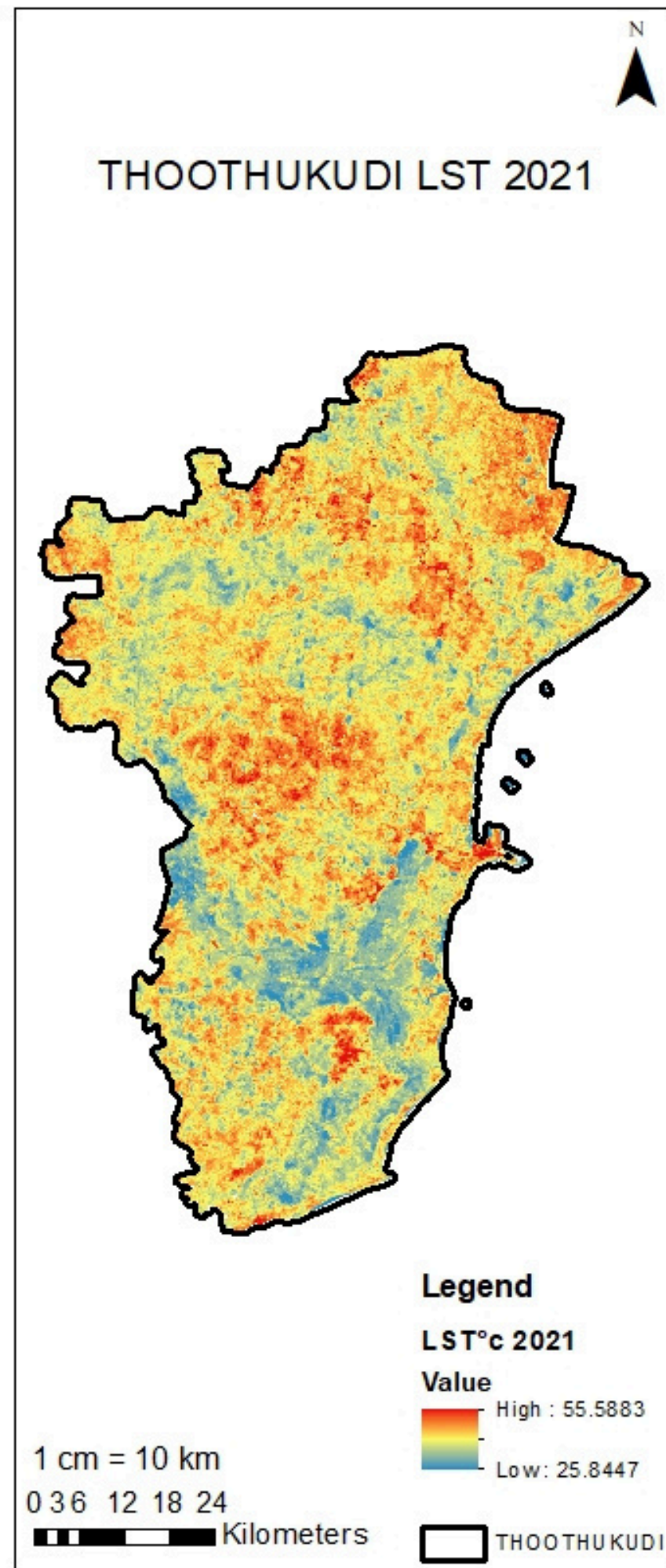
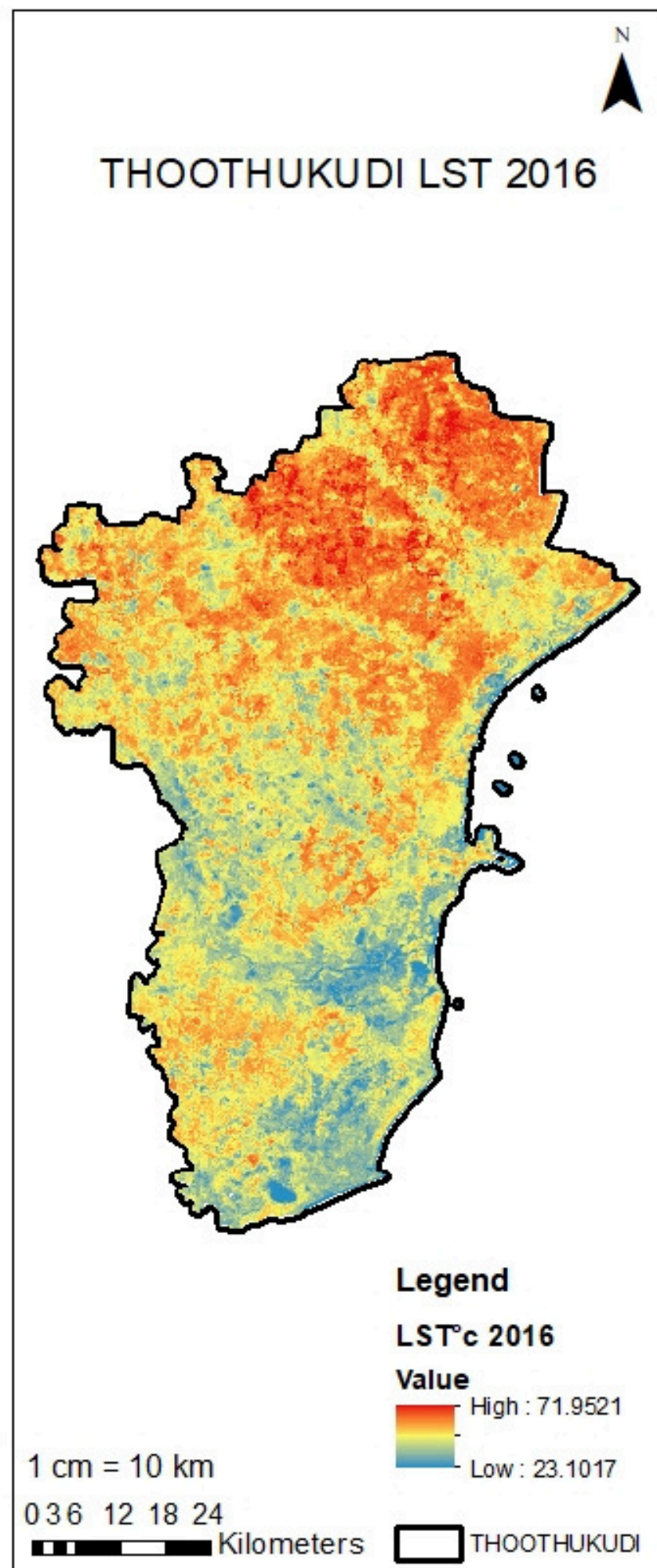
Problem: Salt pans and industrial expansion are erasing natural coastal heat buffers.

Impact: The "Gray" (concrete) is winning over the "Green" (vegetation).

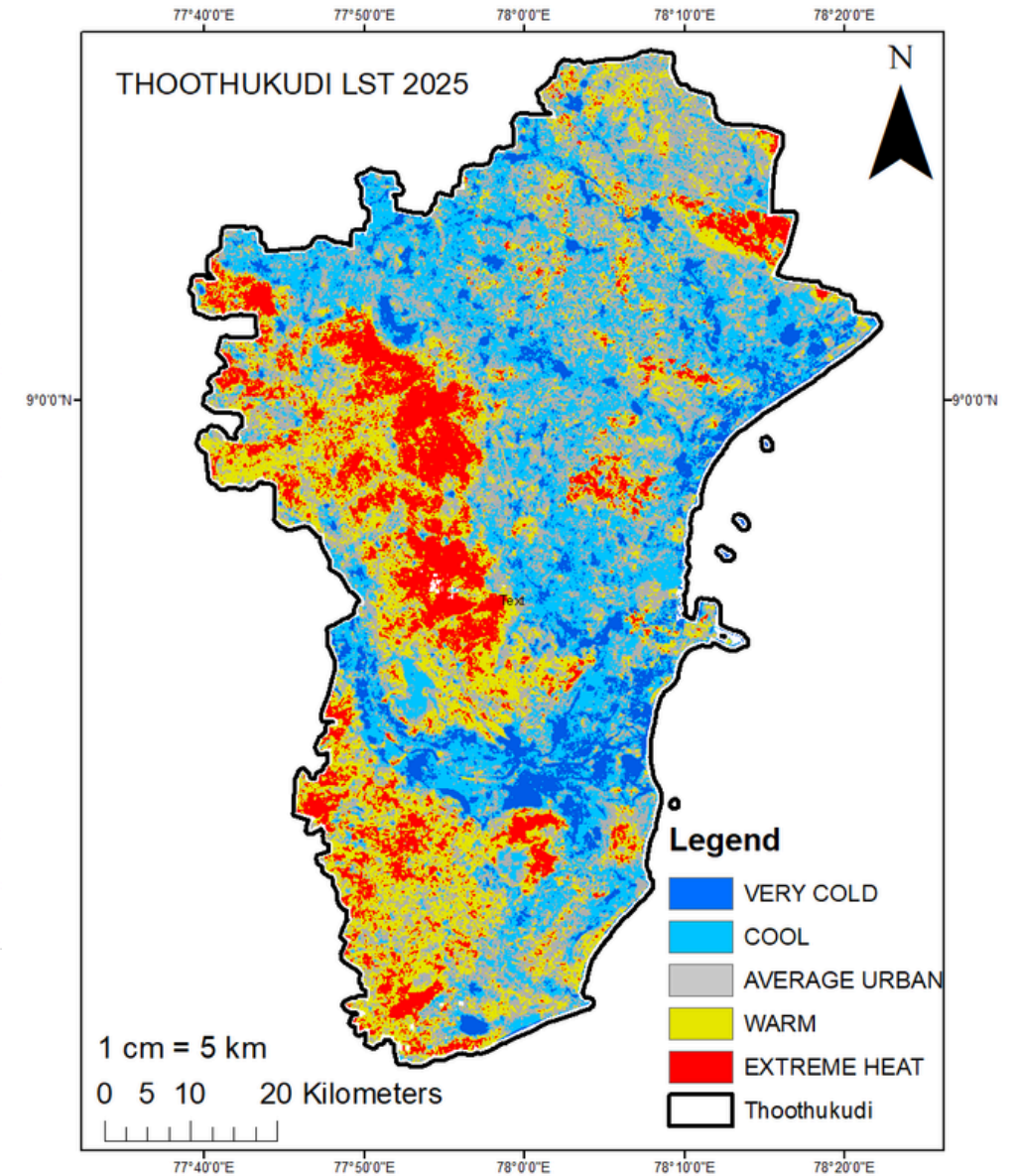
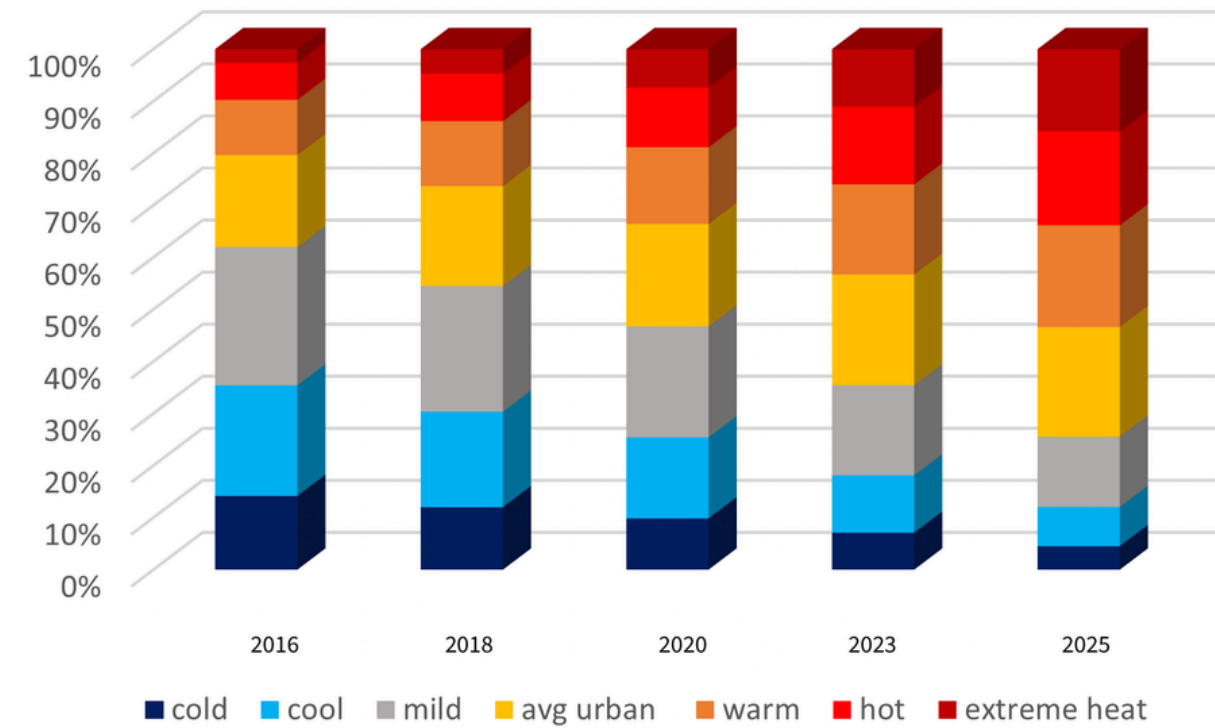
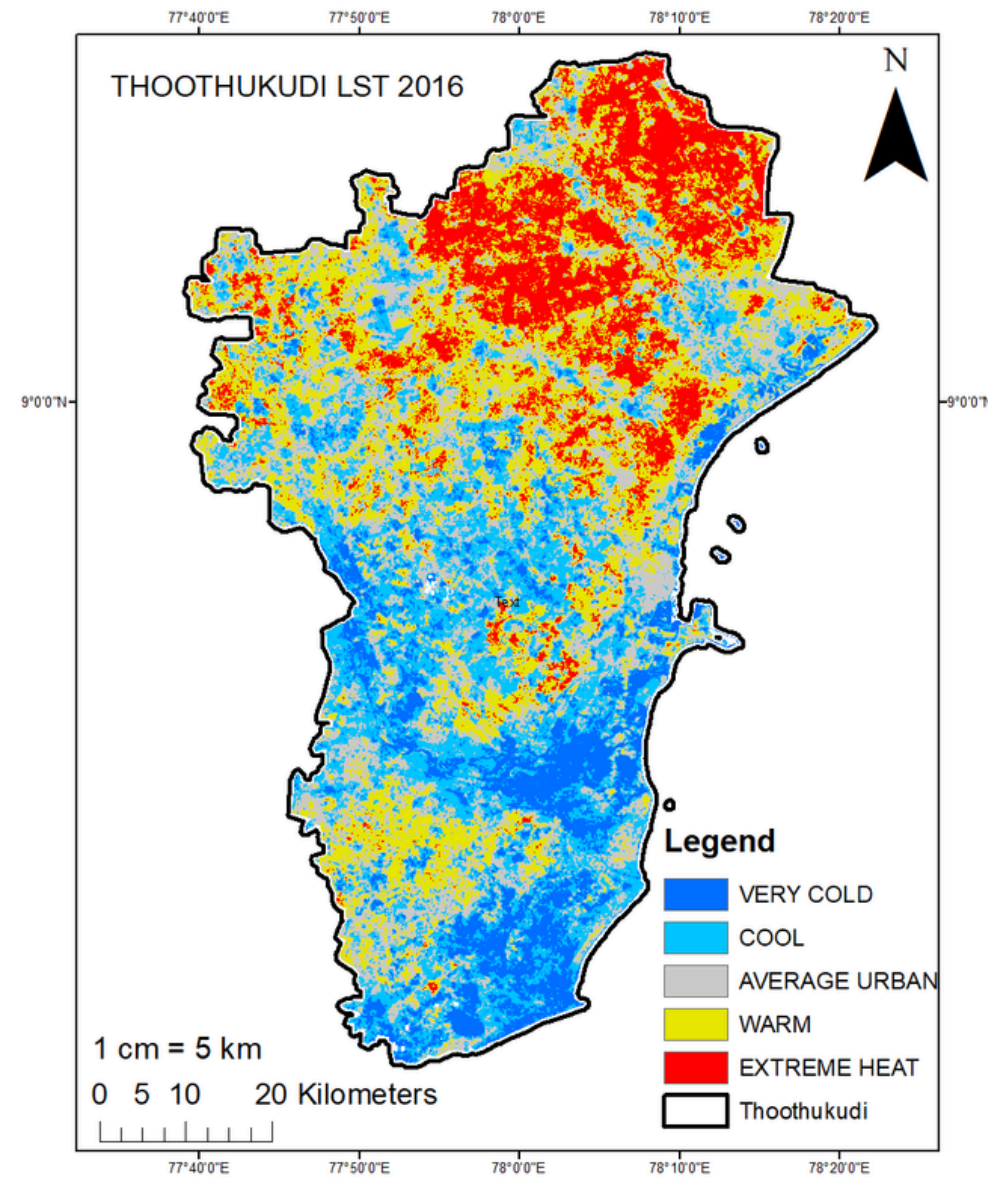
Health & Sustainability at Risk

Problem: Extreme heat threatens outdoor workers and spikes energy demand.

Impact: A "Smart City" isn't smart if it's too hot to live in.



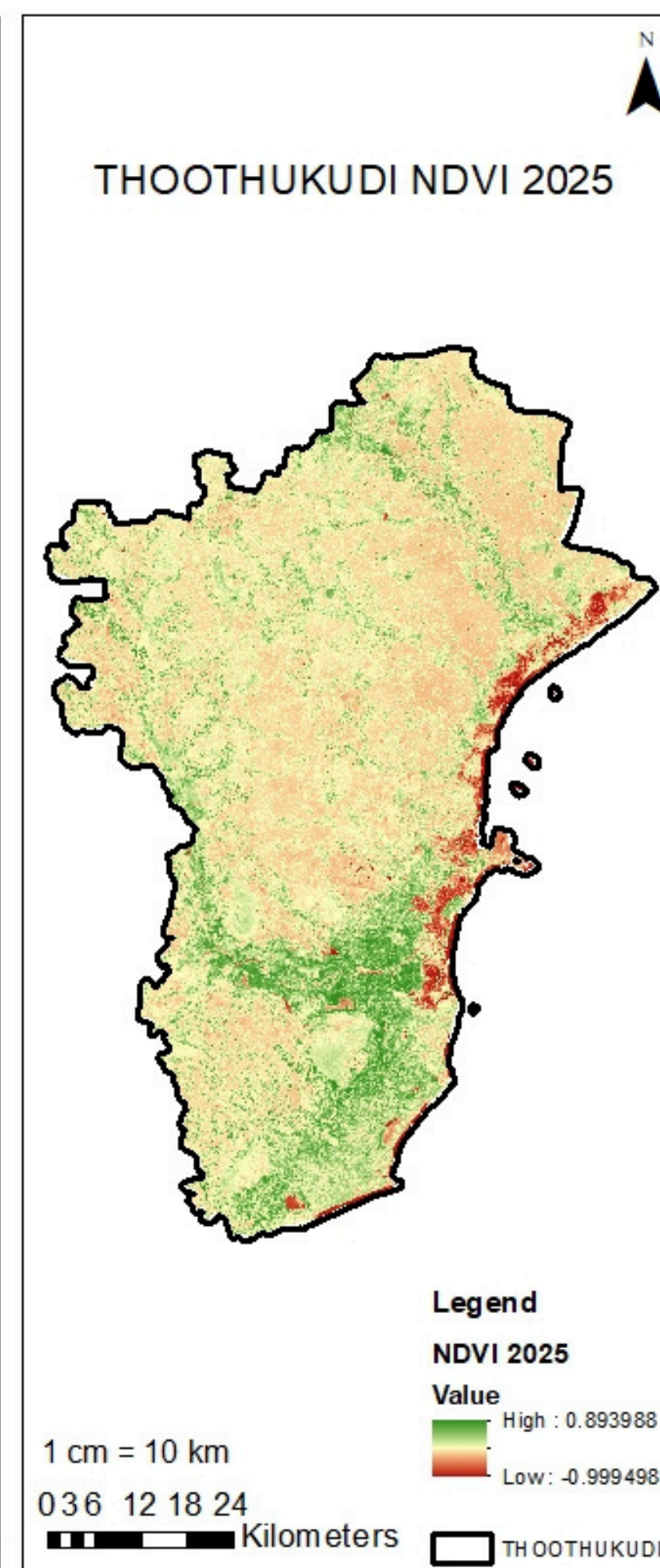
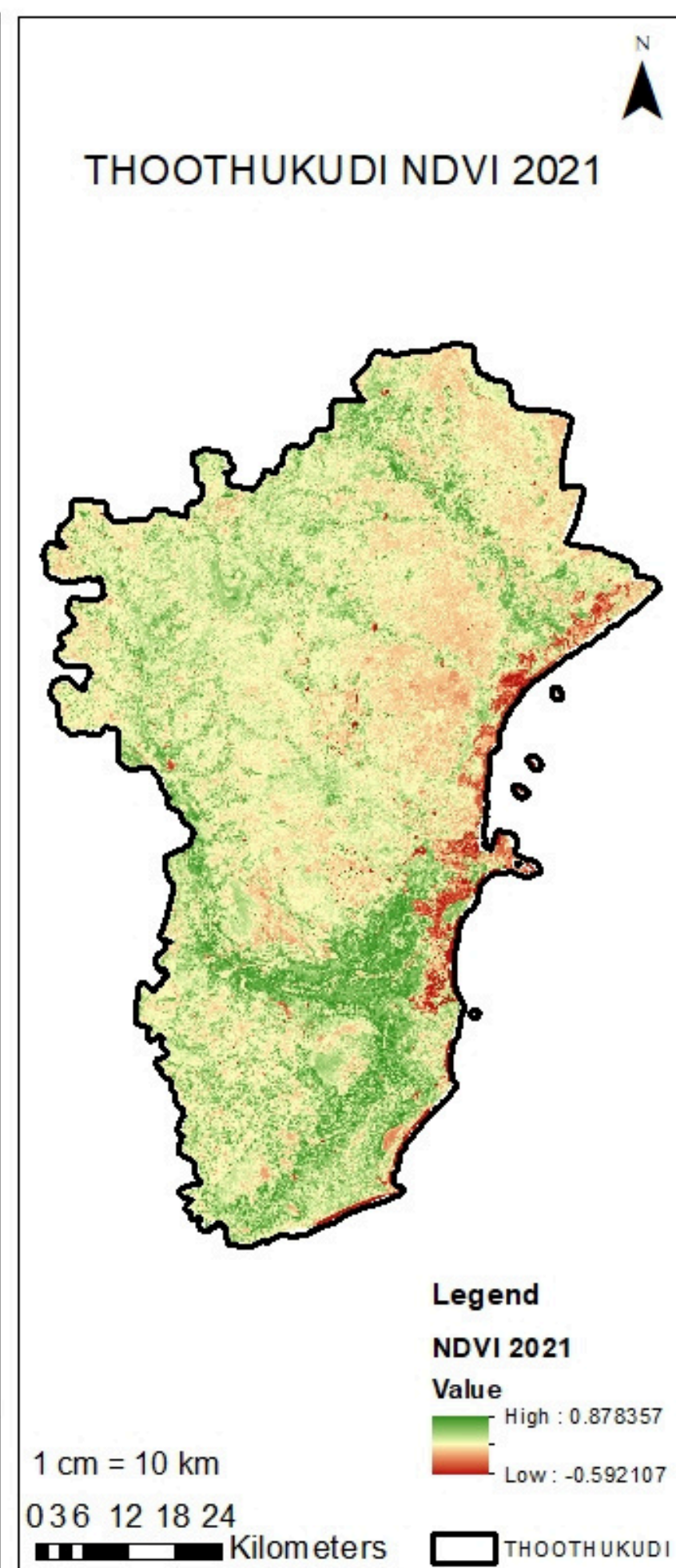
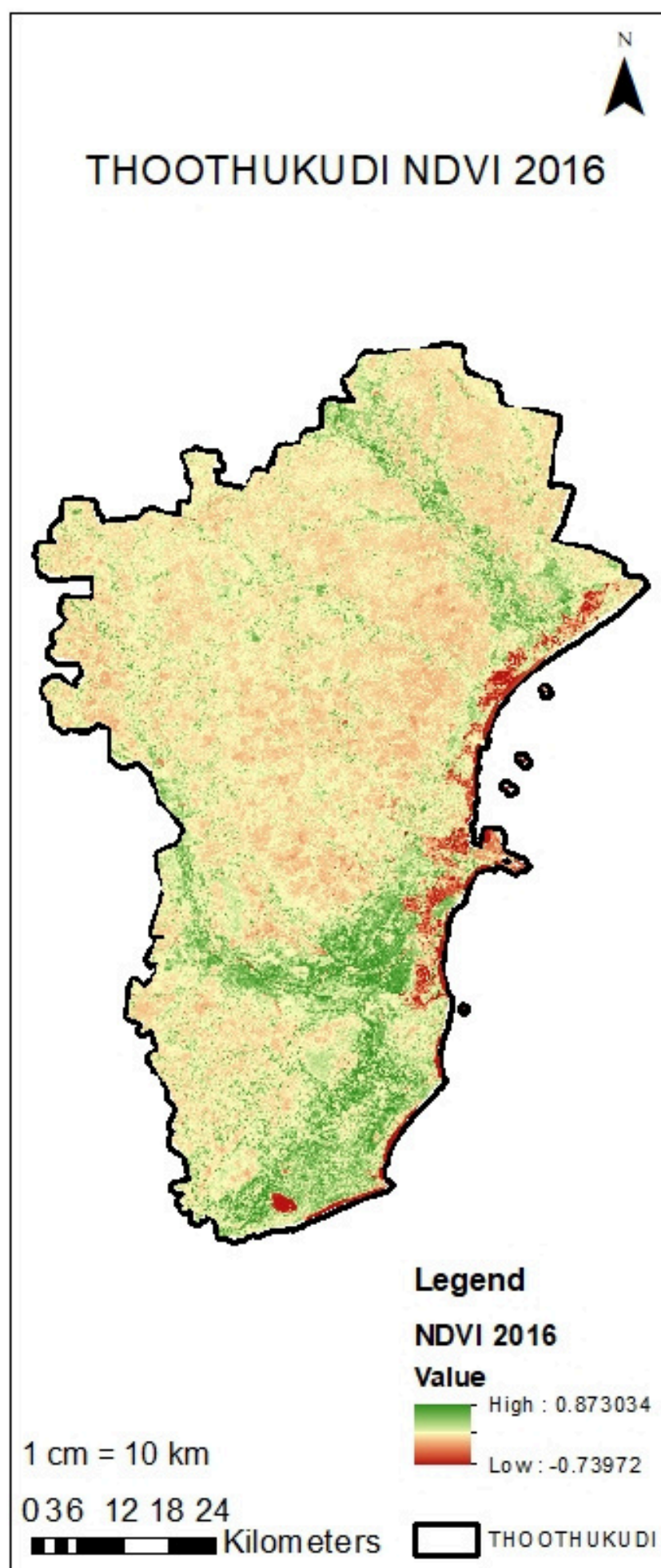
LAND SURFACE TEMPERATURE



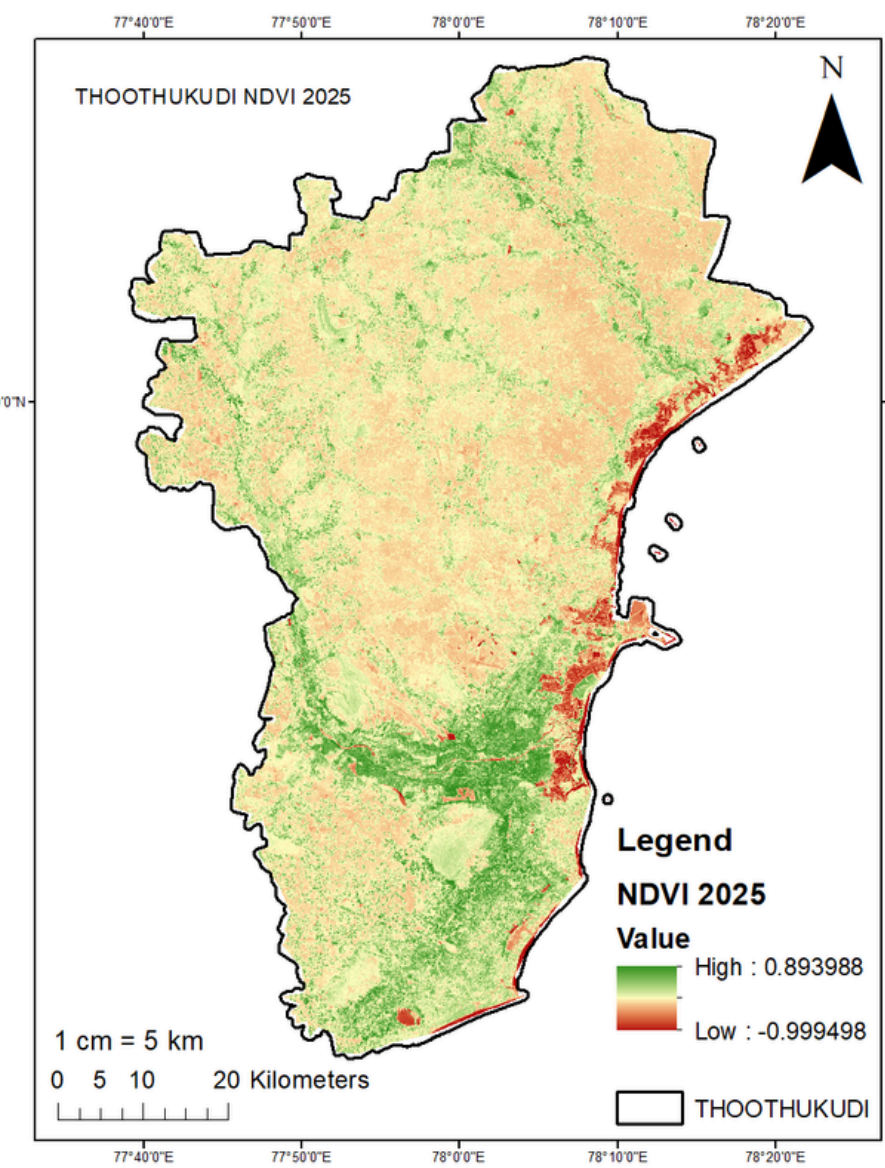
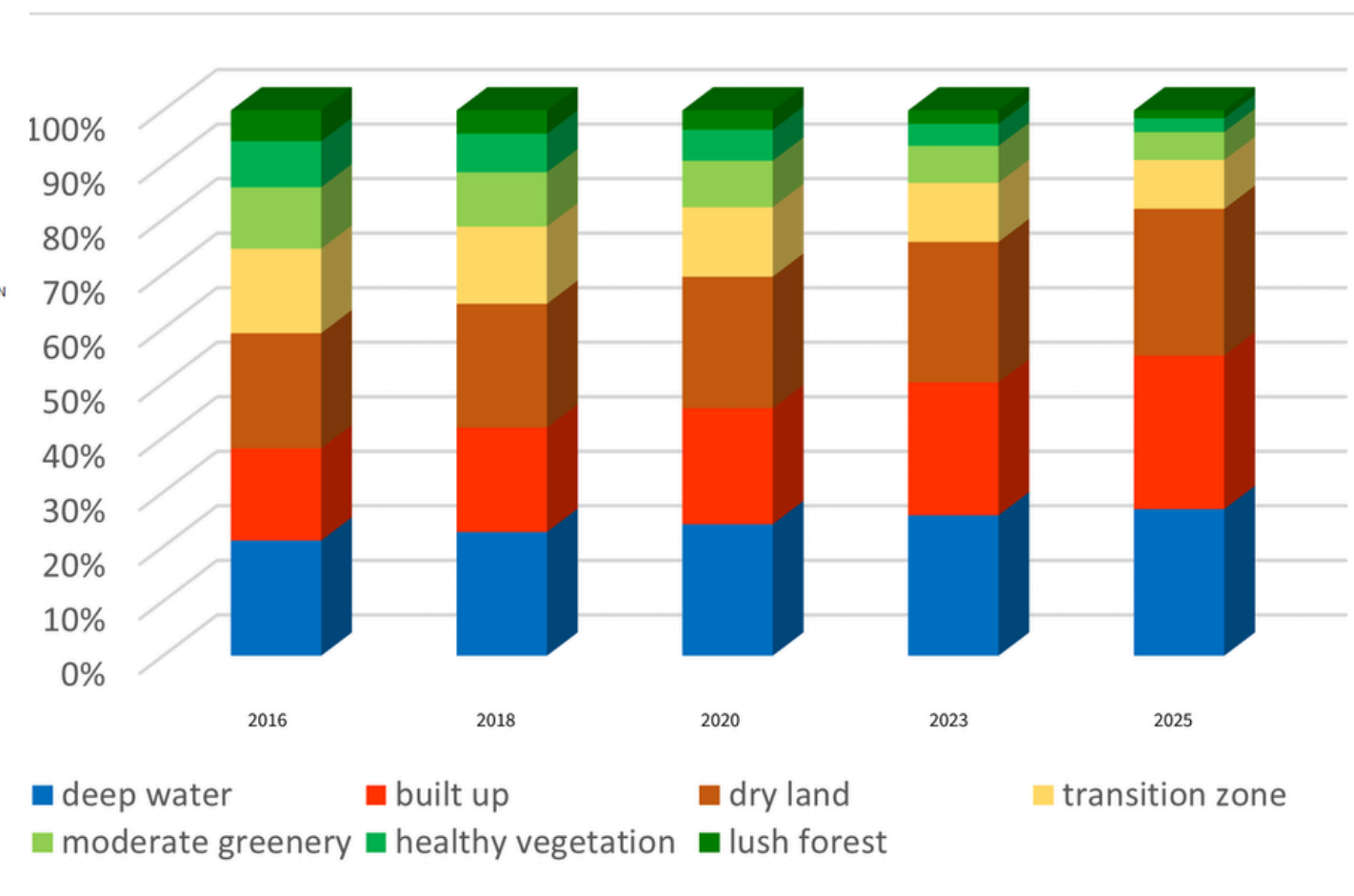
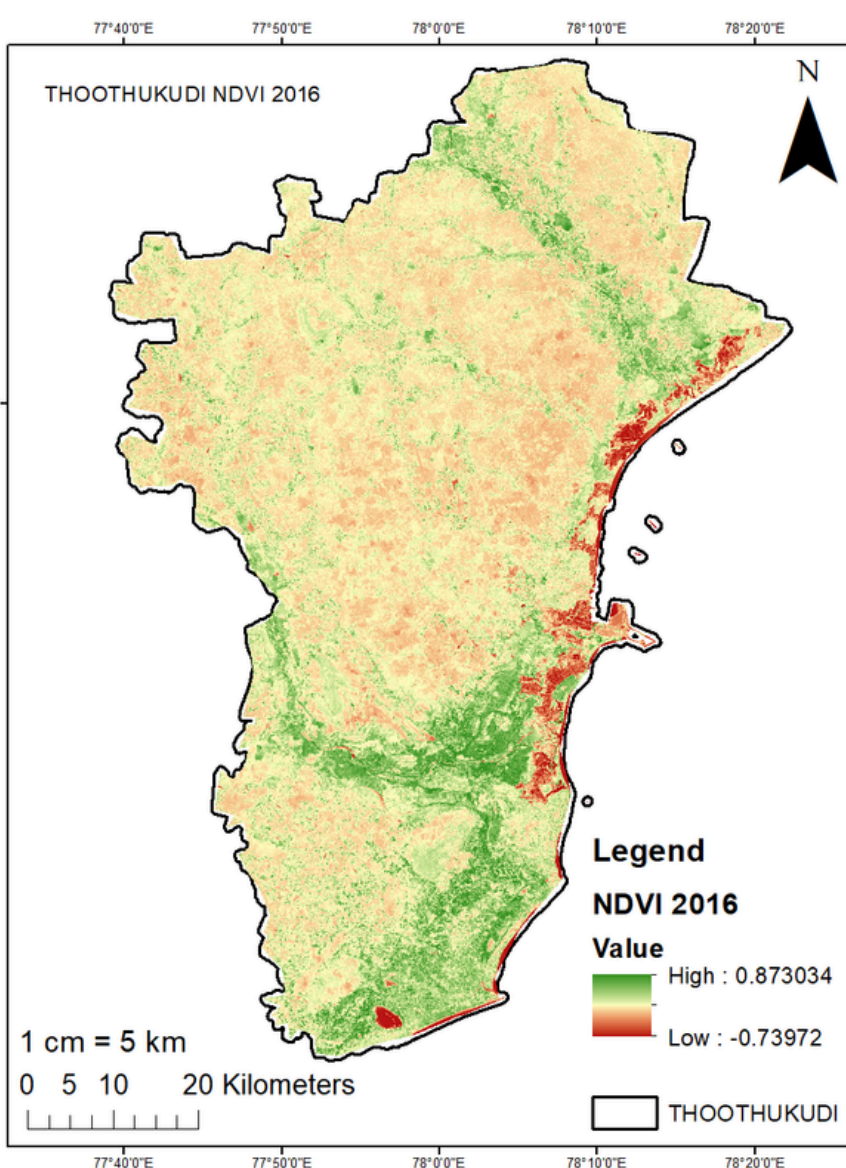
Rising Land Surface Temperature (LST)

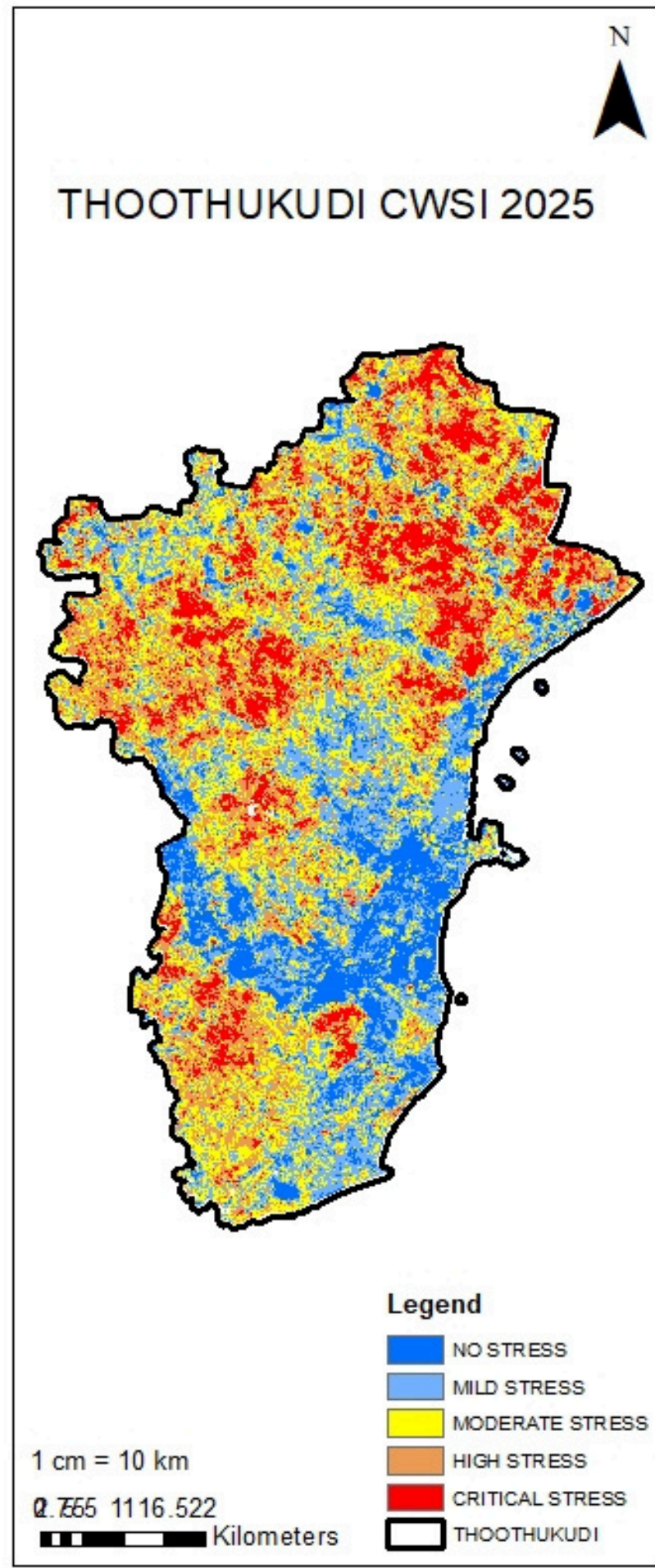
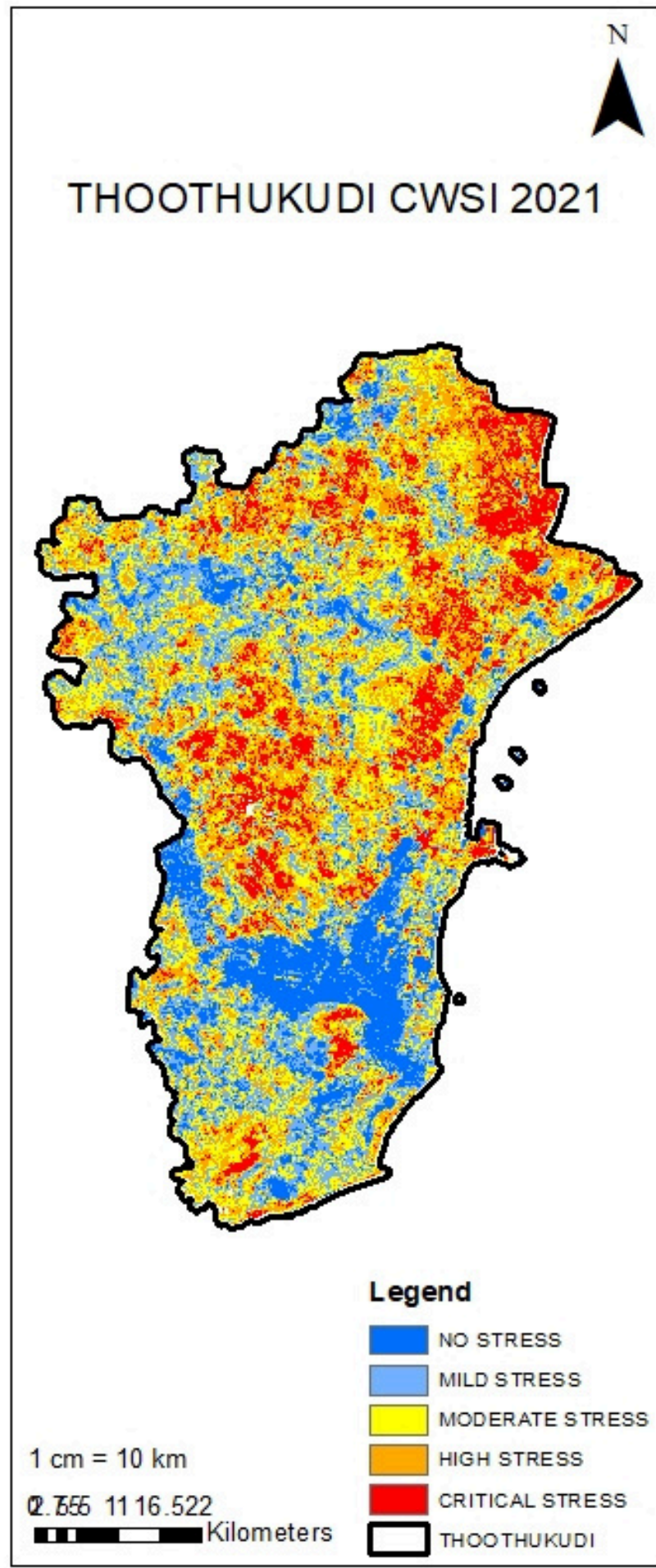
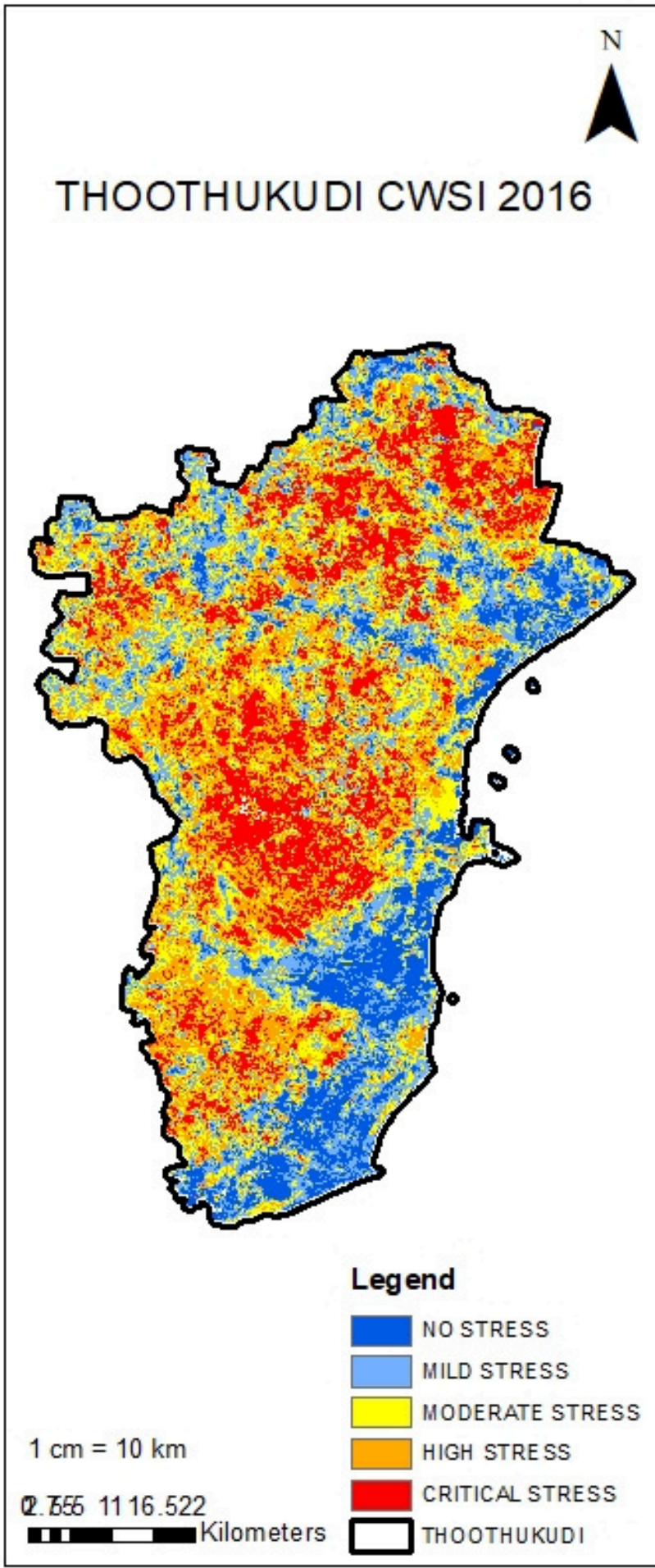
- 2016: Temperatures were moderate, with fewer extreme red zones.
- 2021: Some cooling anomalies appear, but overall warming trends persist.
- 2025: Sharp increase in maximum LST (82.9°C), with widespread red/orange zones.

Implication: High LST directly stresses vegetation (NDVI decline) and intensifies Urban Heat Island (UHI). It also raises risks for human health (heat stress, dehydration, cardiovascular strain).

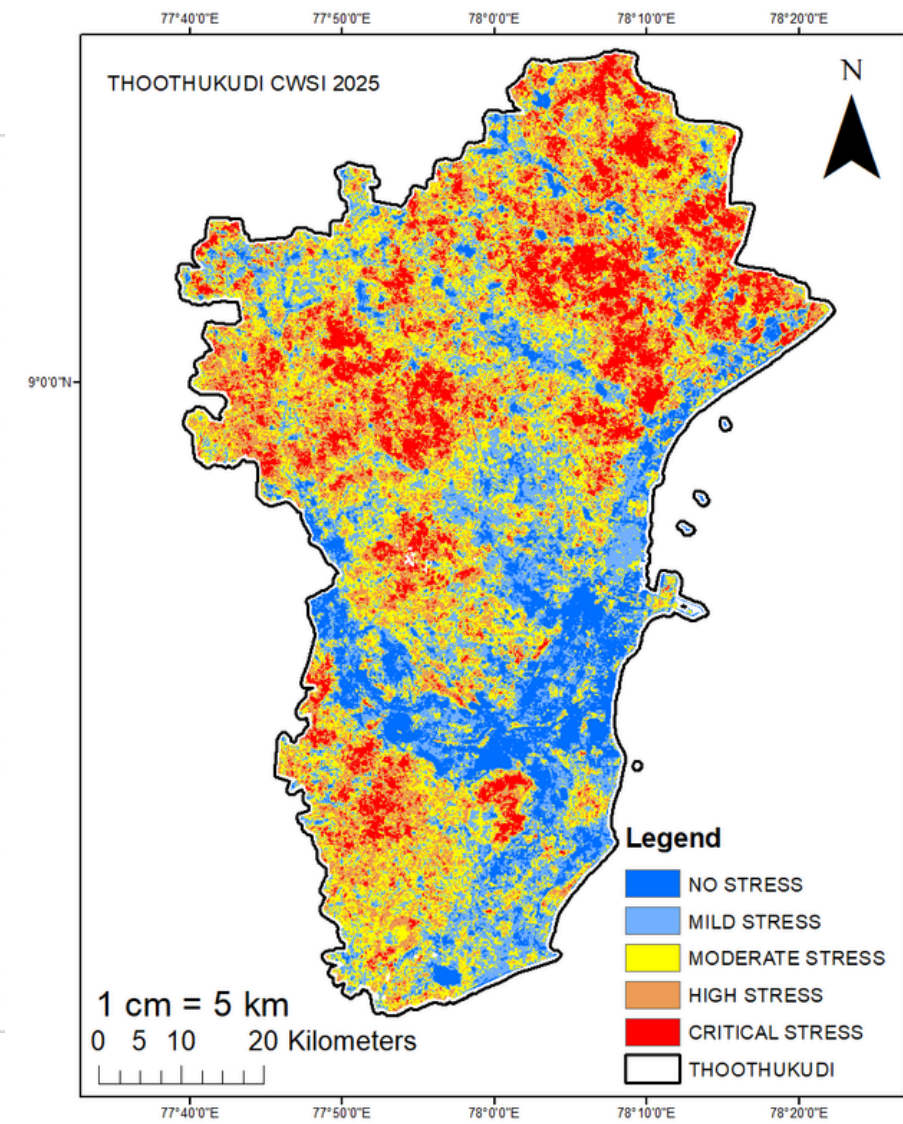
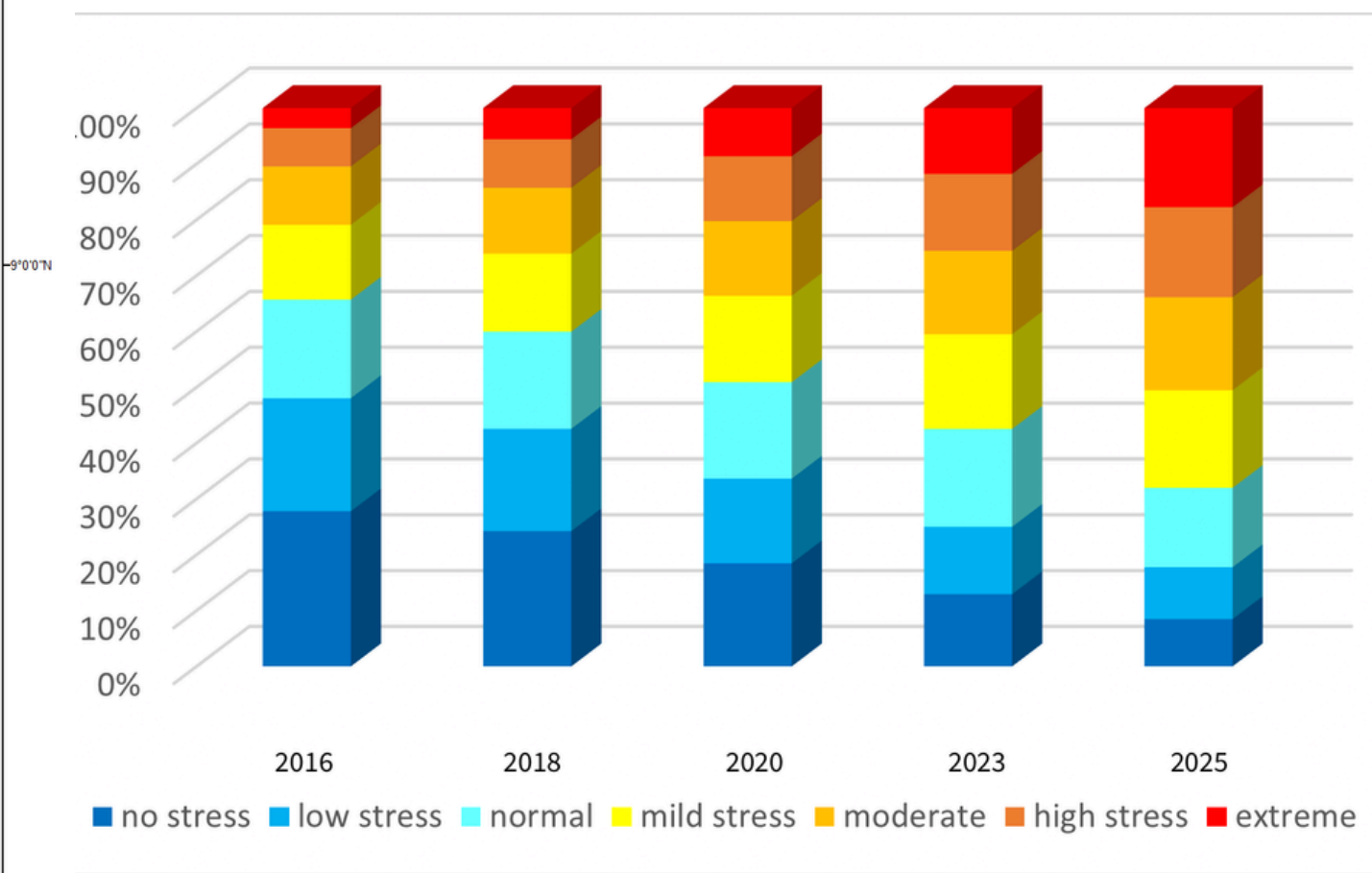
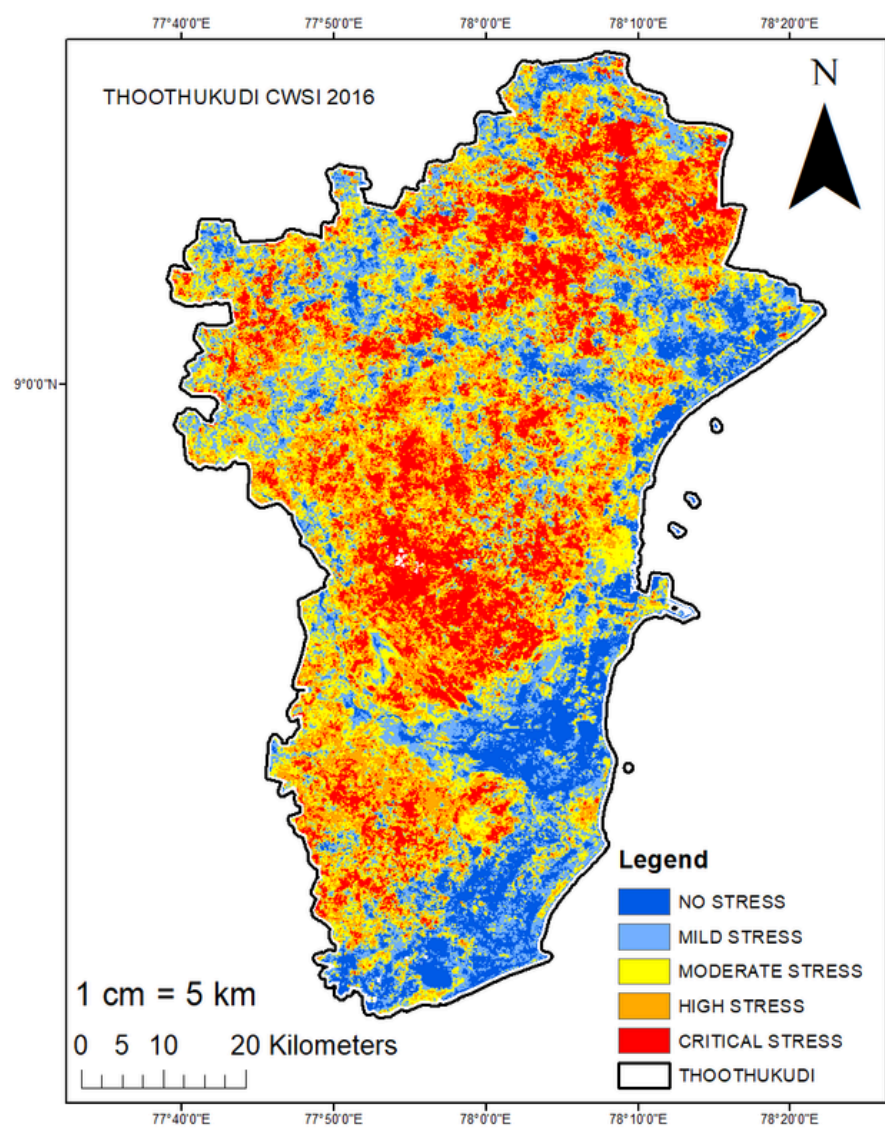


NORMAL DIFFERENCE VEGETATION INDEX





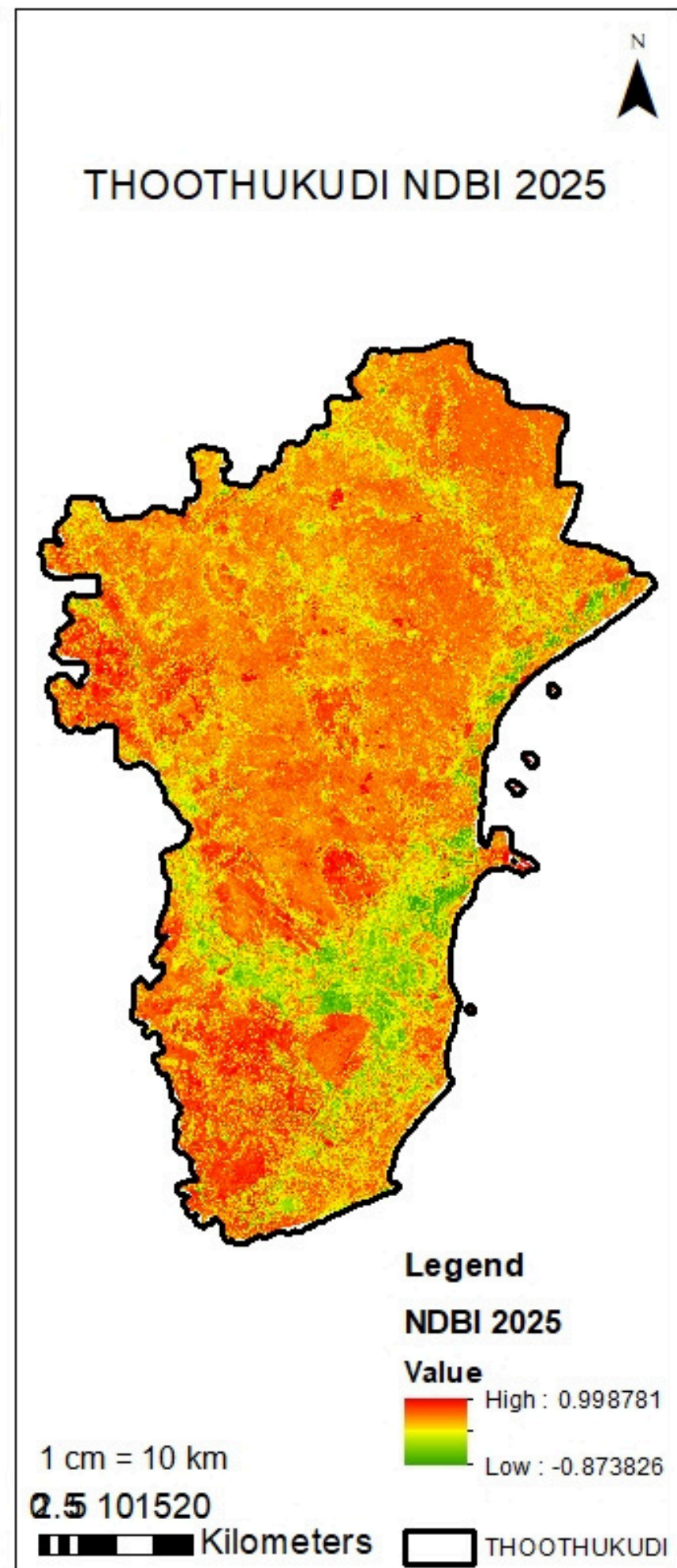
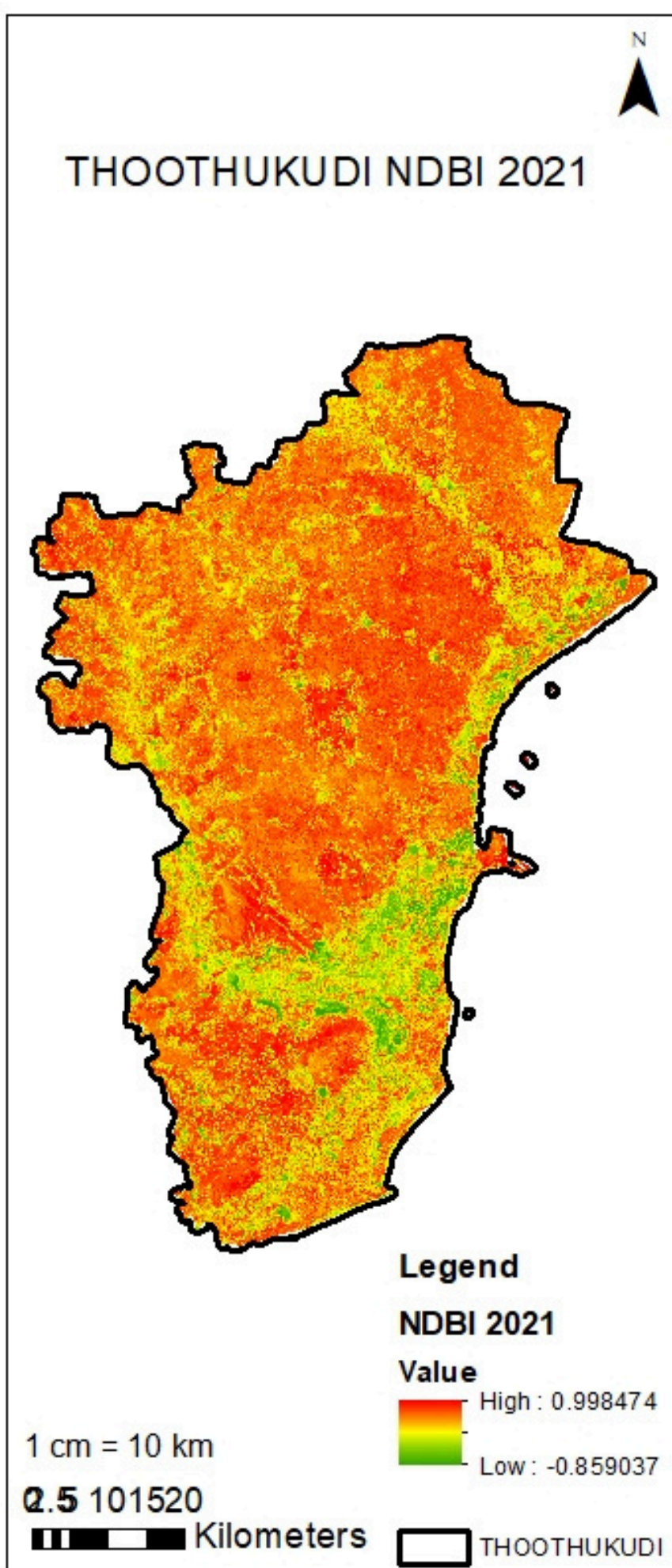
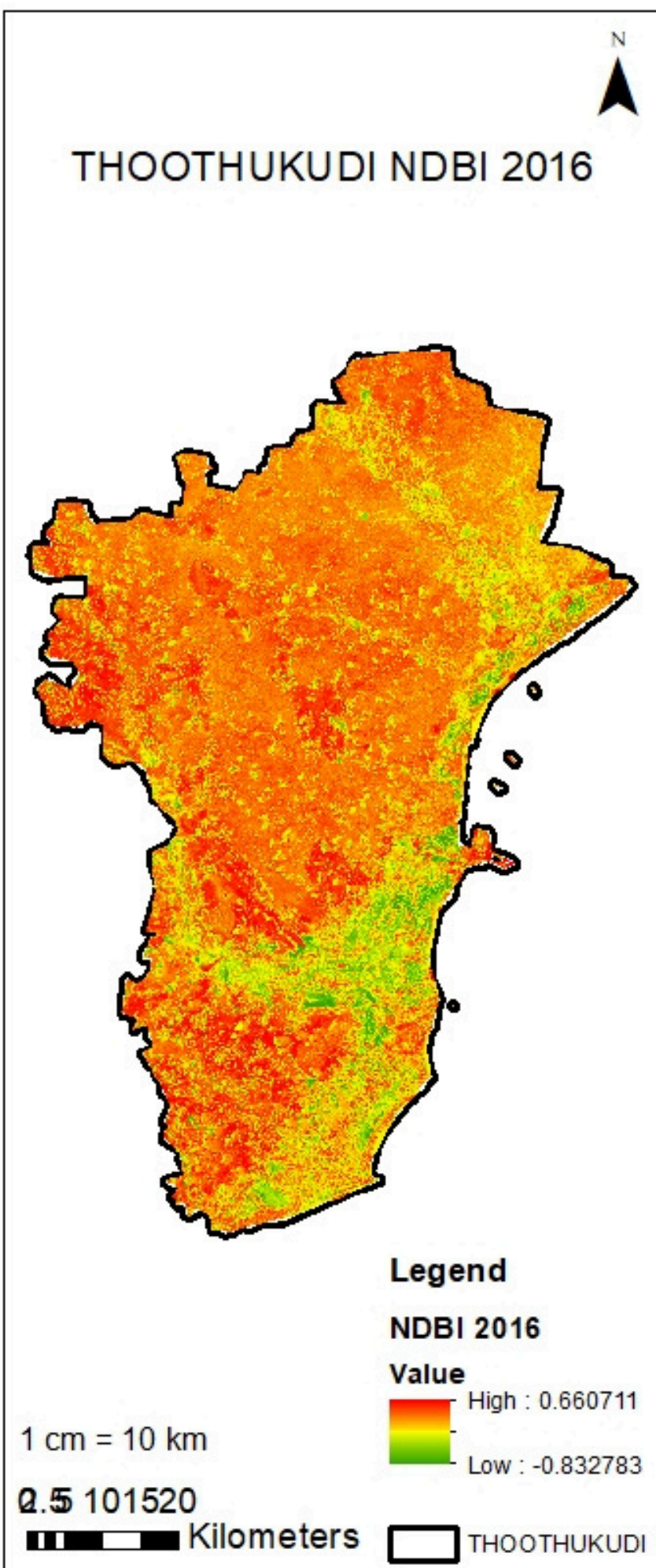
CROP WATER STRESS INDEX



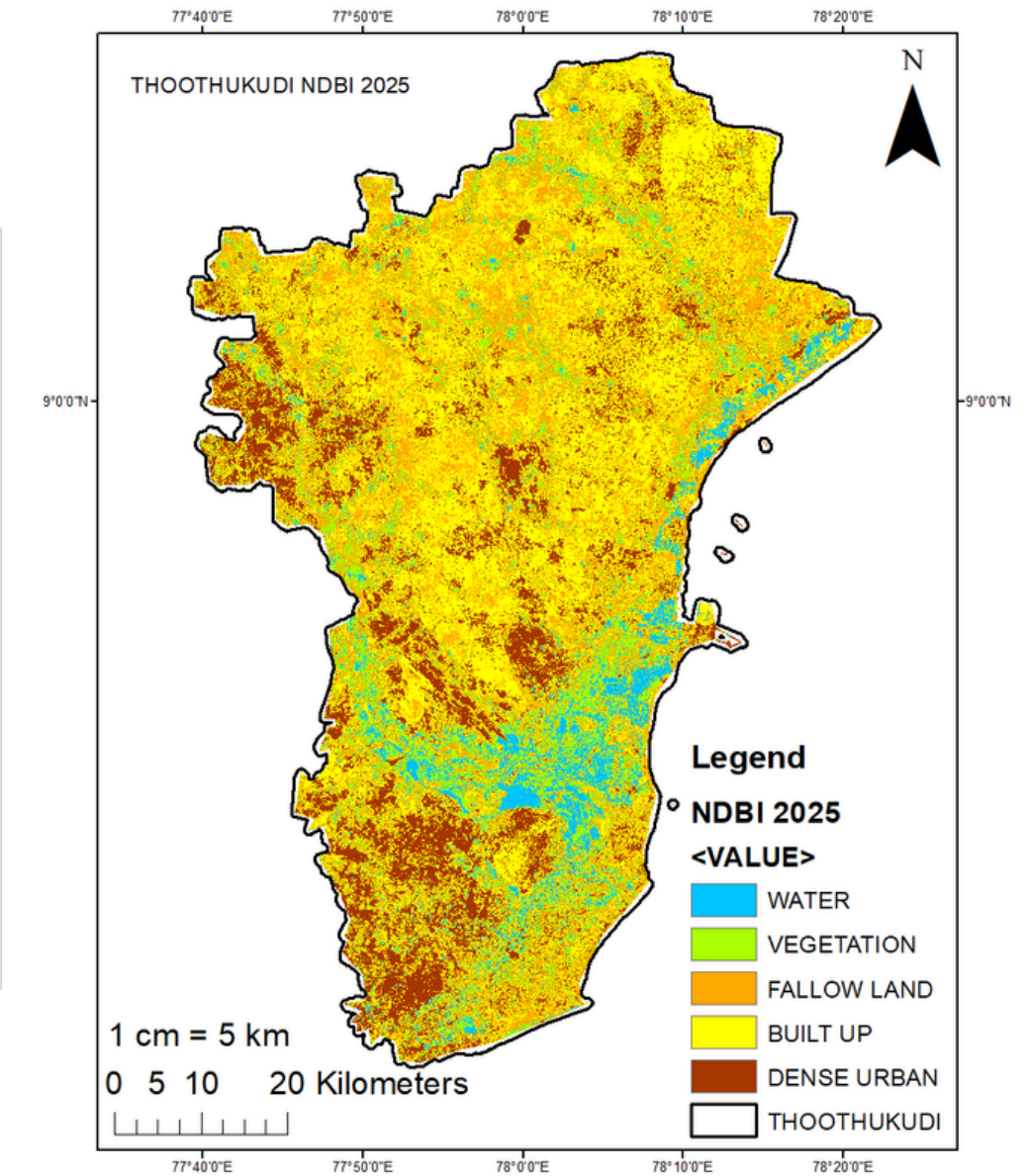
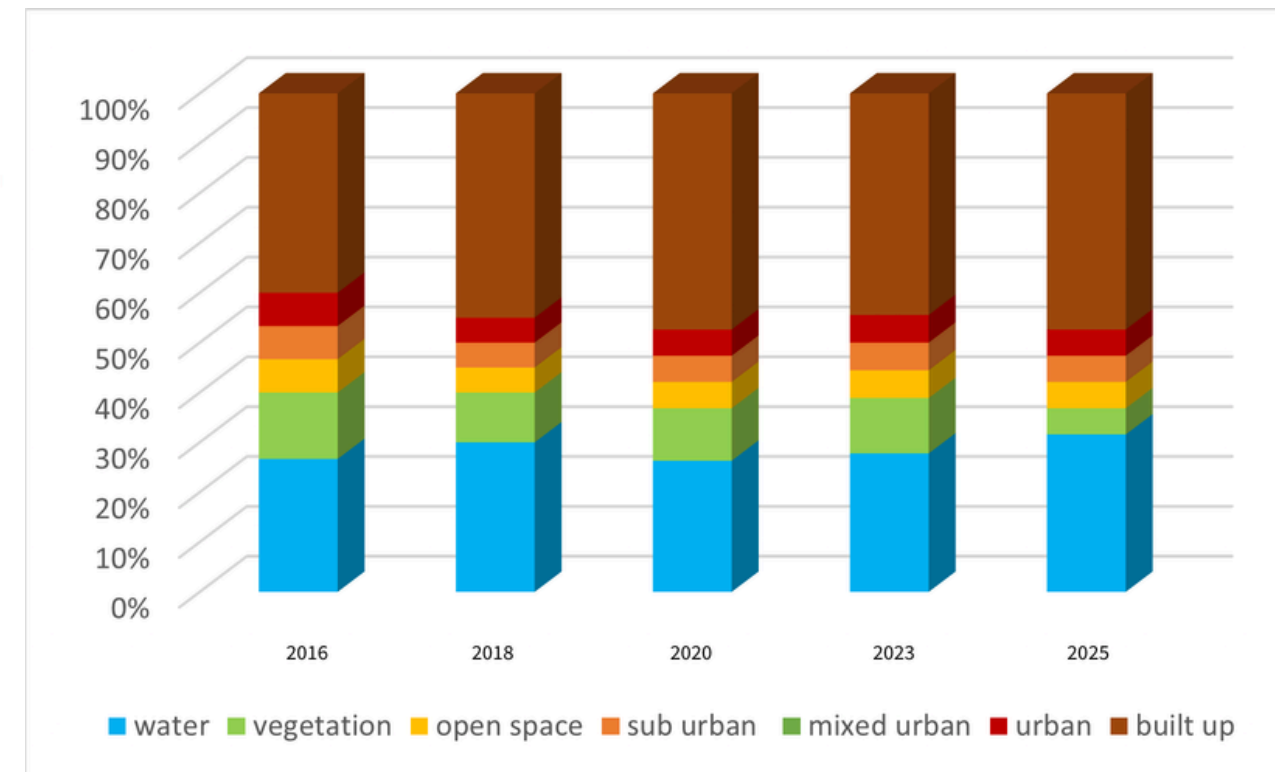
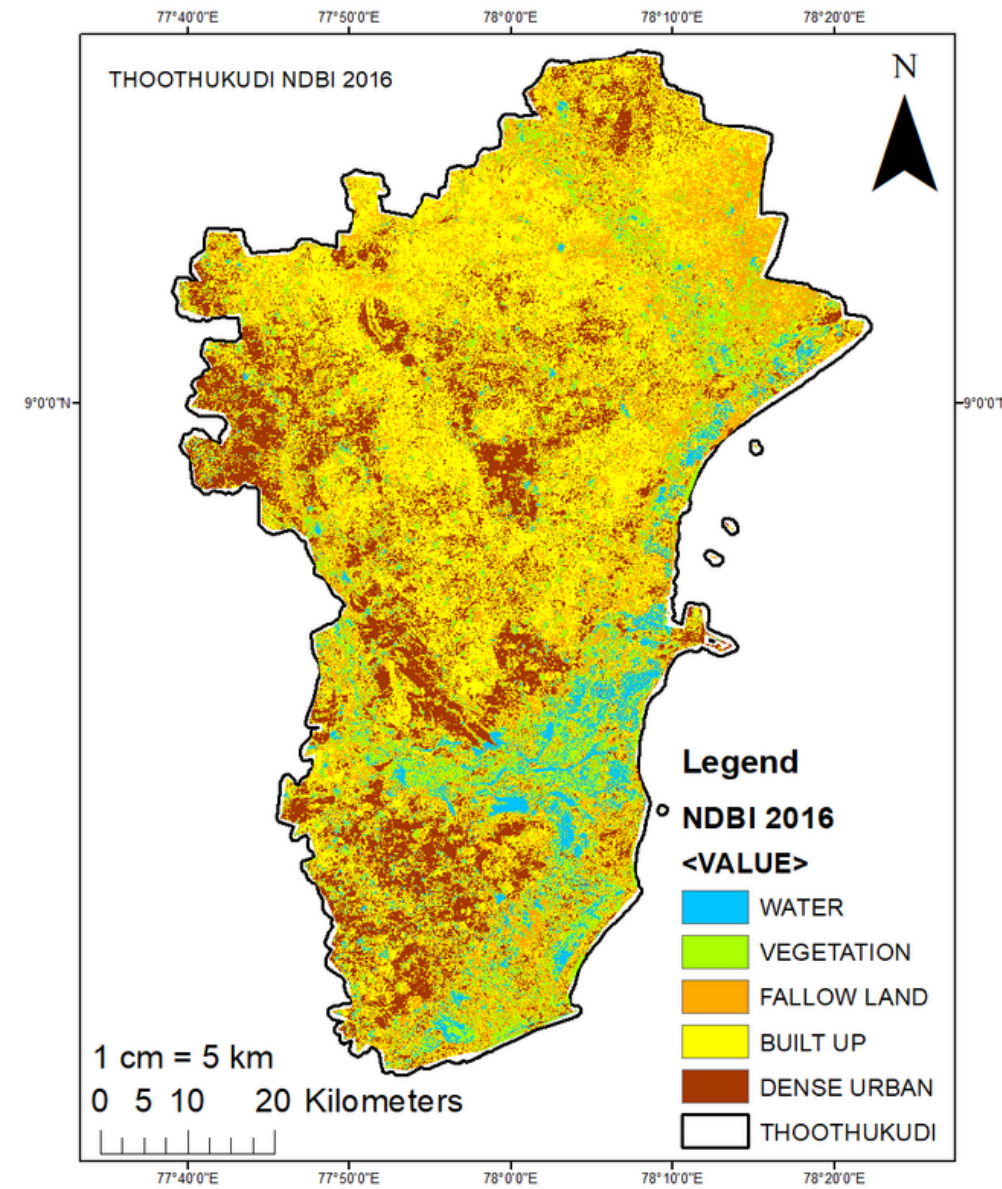
Vegetation Stress (NDVI & CWSI)

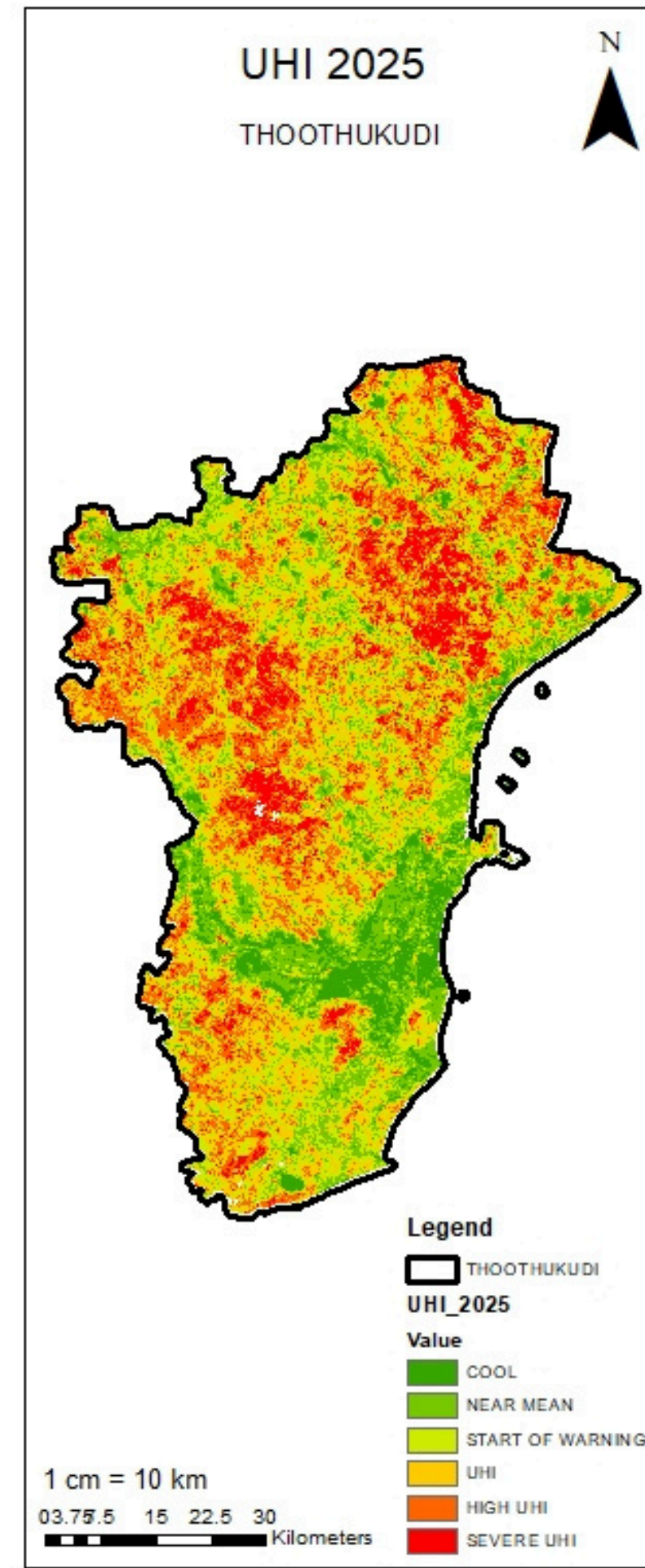
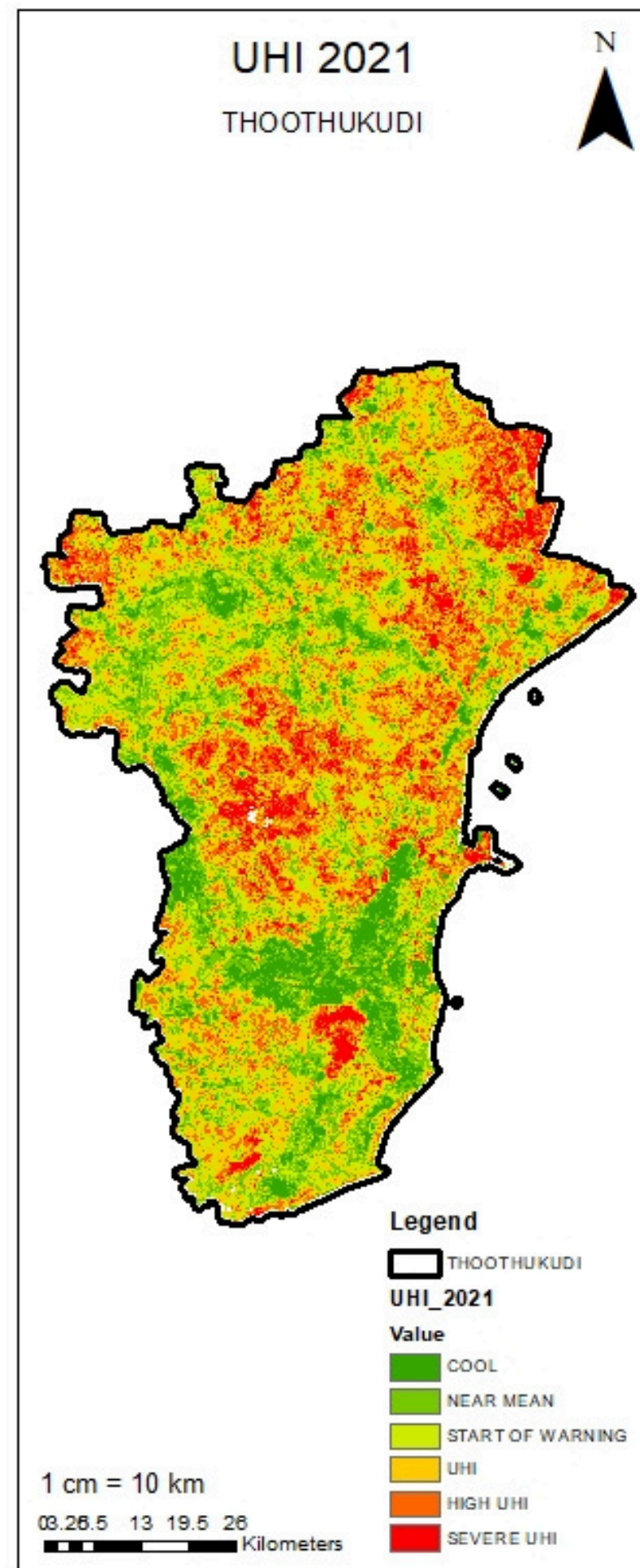
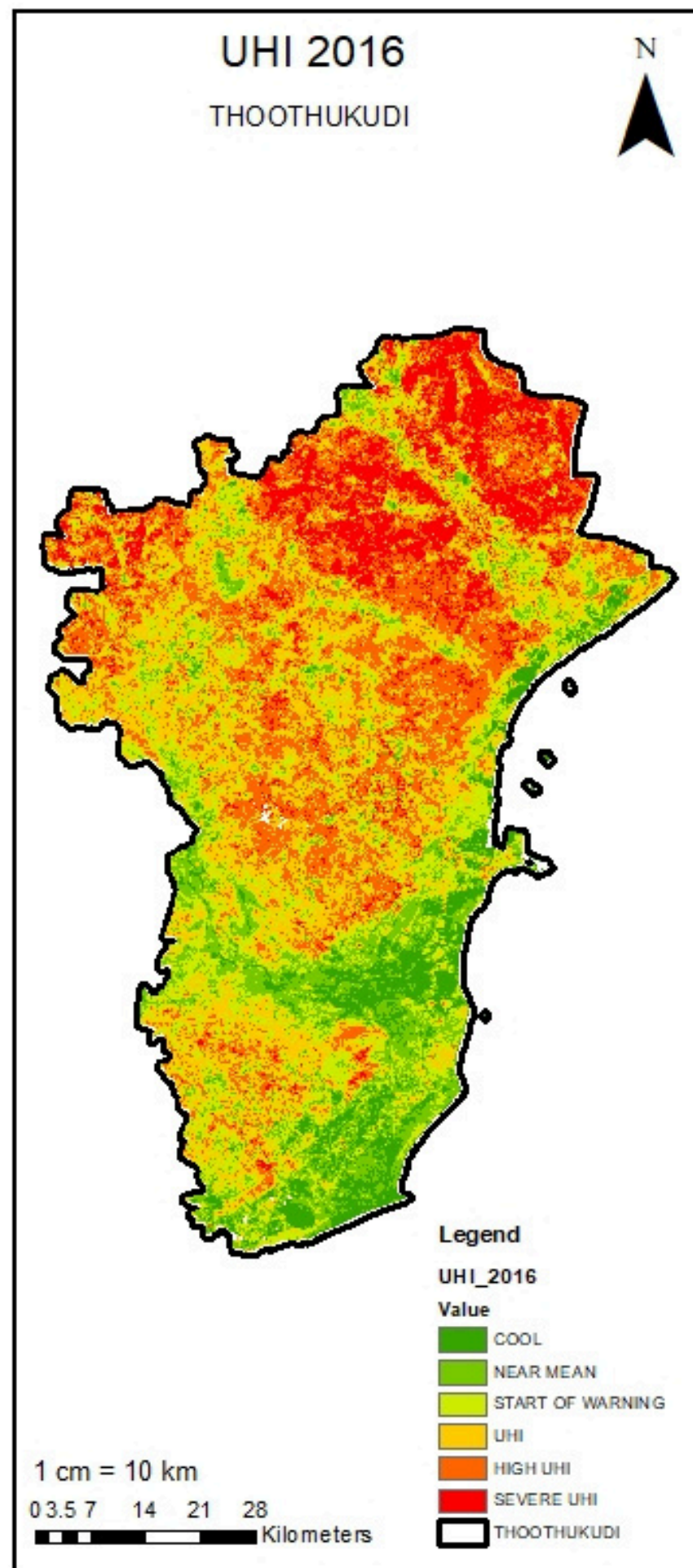
- NDVI:
 - 2016: More green zones (healthy vegetation).
 - 2021: Noticeable decline in dense vegetation, more red patches.
 - 2025: Severe degradation, with large areas showing negative NDVI values.
- CWSI (Crop Water Stress Index):
 - 2016: Mostly mild/moderate stress.
 - 2021: Expansion of high stress zones.
 - 2025: Critical stress dominates, indicating water scarcity and crop failure risk.

Implication: Rising temperature + reduced vegetation cover = ecological imbalance. Agriculture suffers, groundwater demand spikes, and food security weakens.

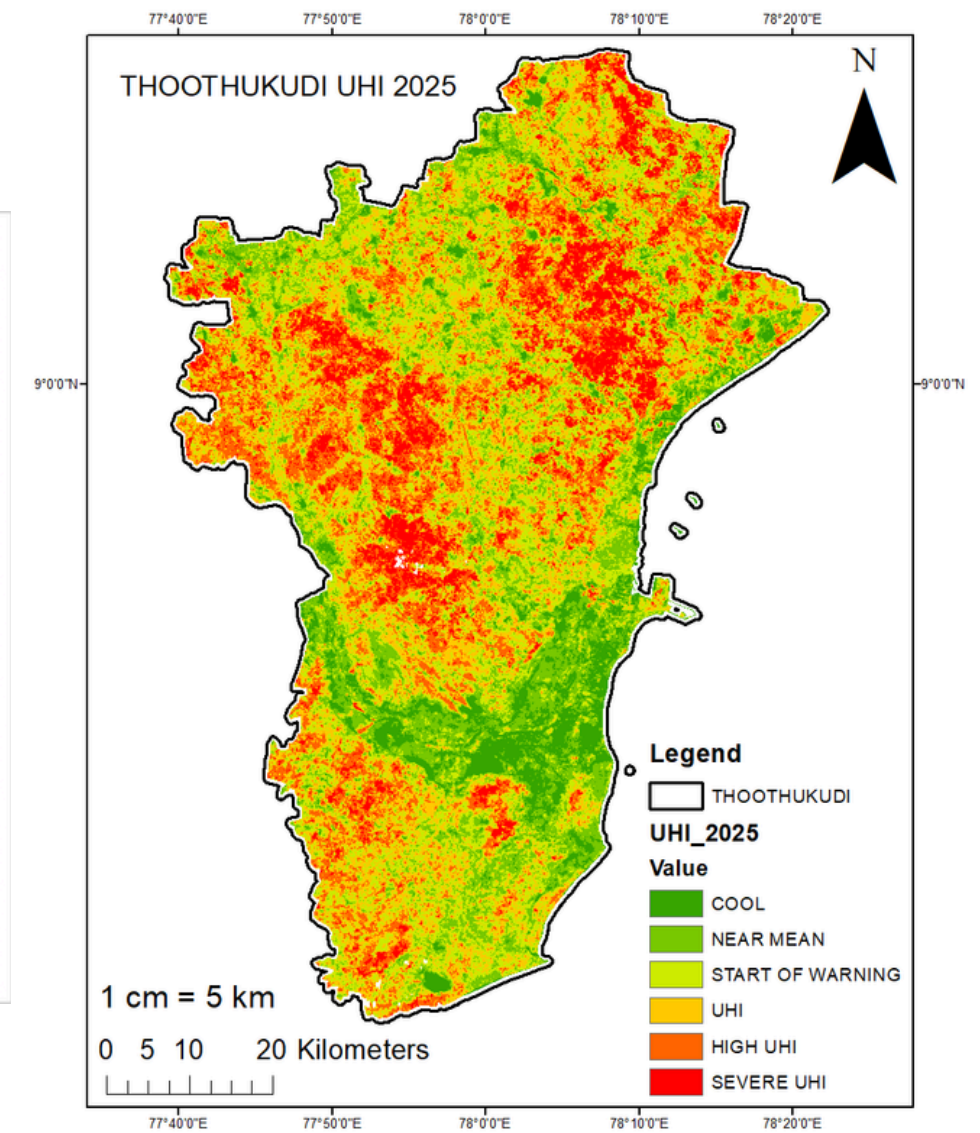
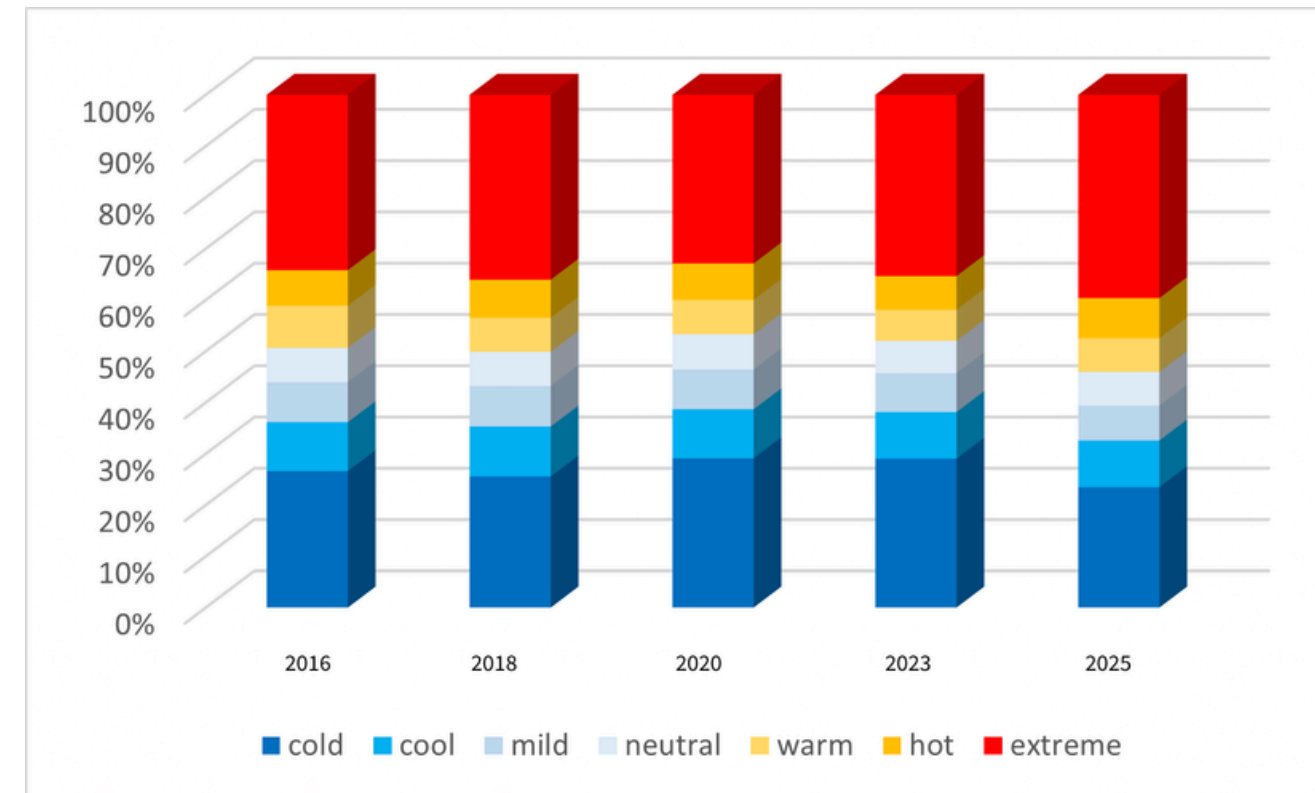
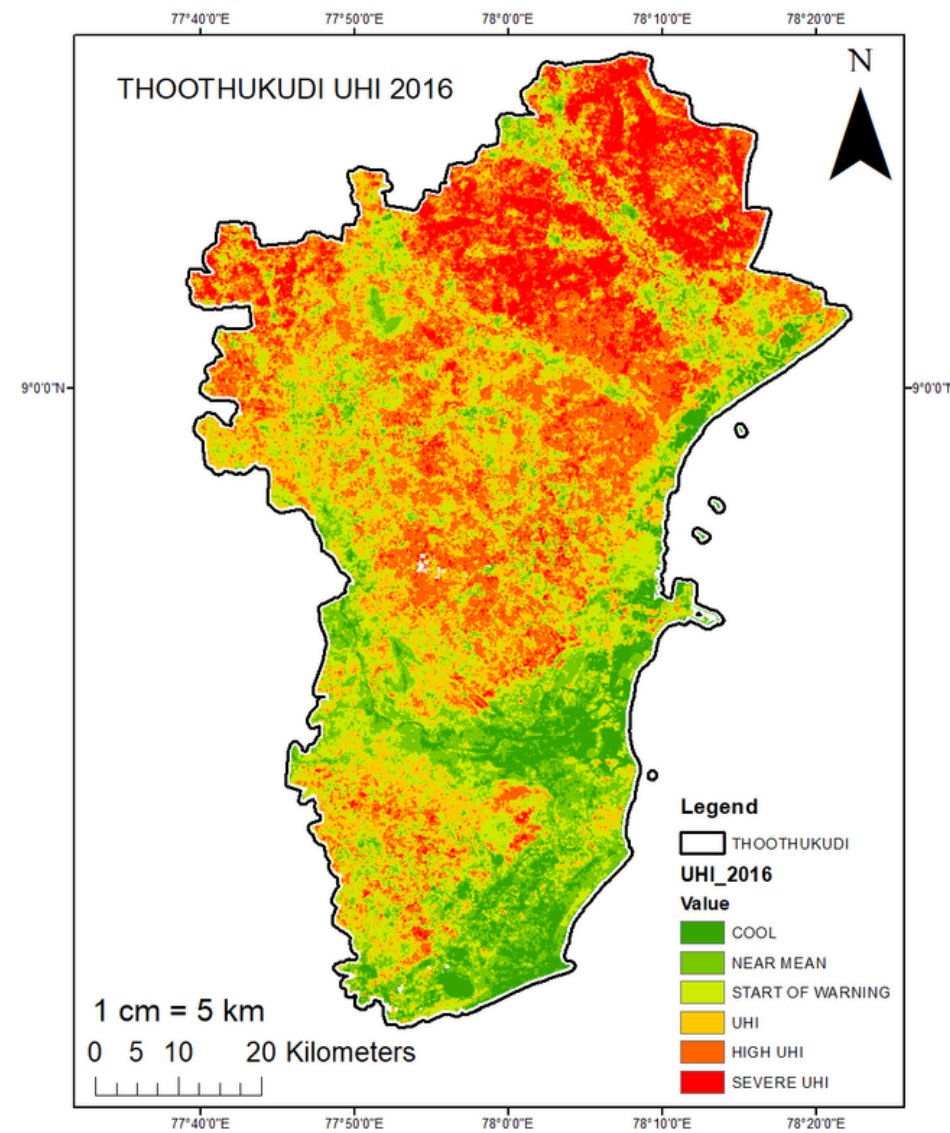


NORMAL DIFFERENCE BUILTUP INDEX





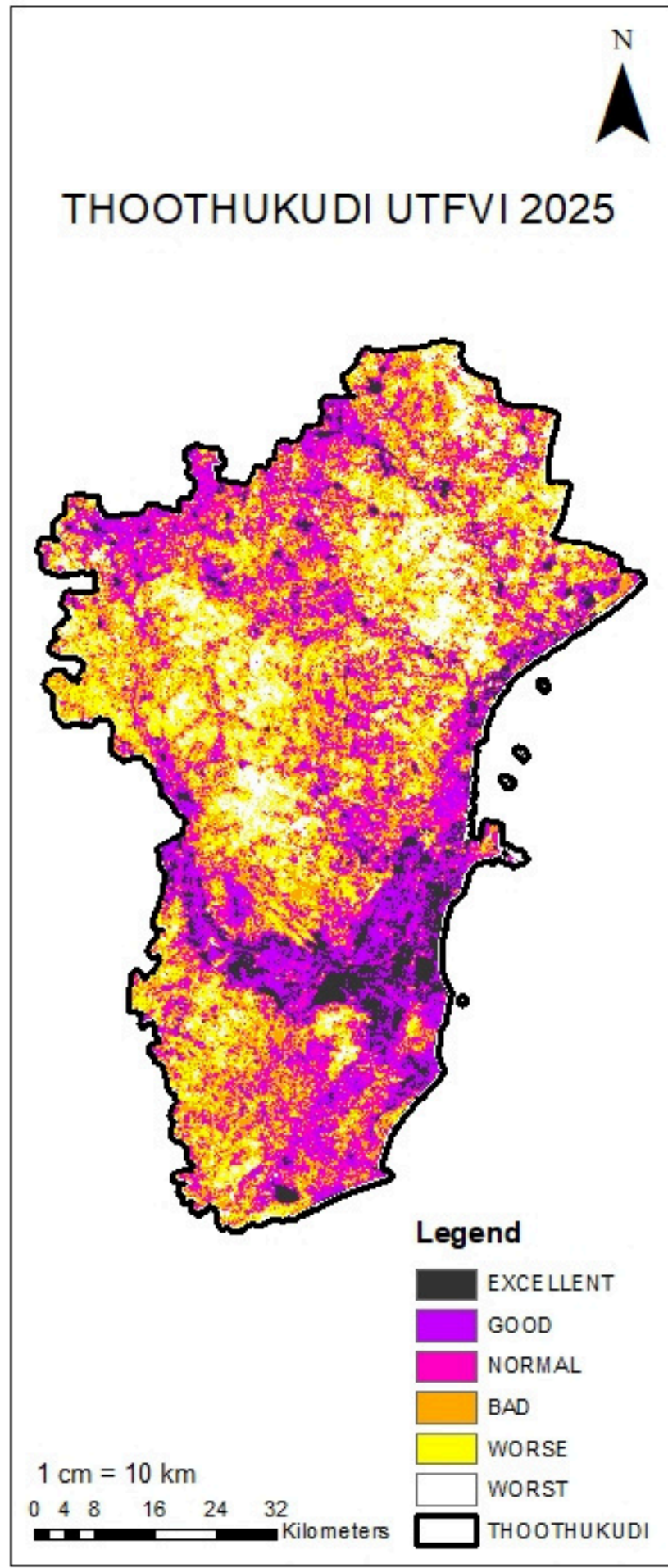
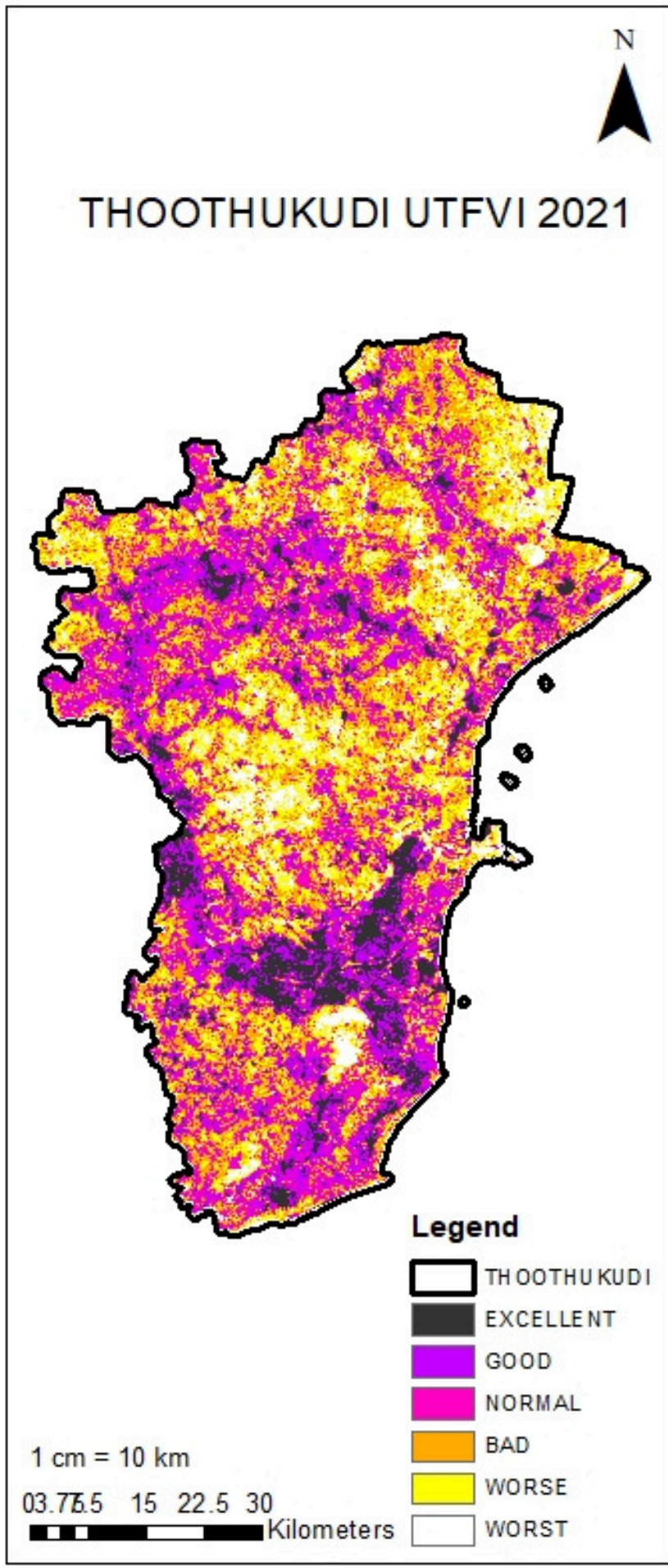
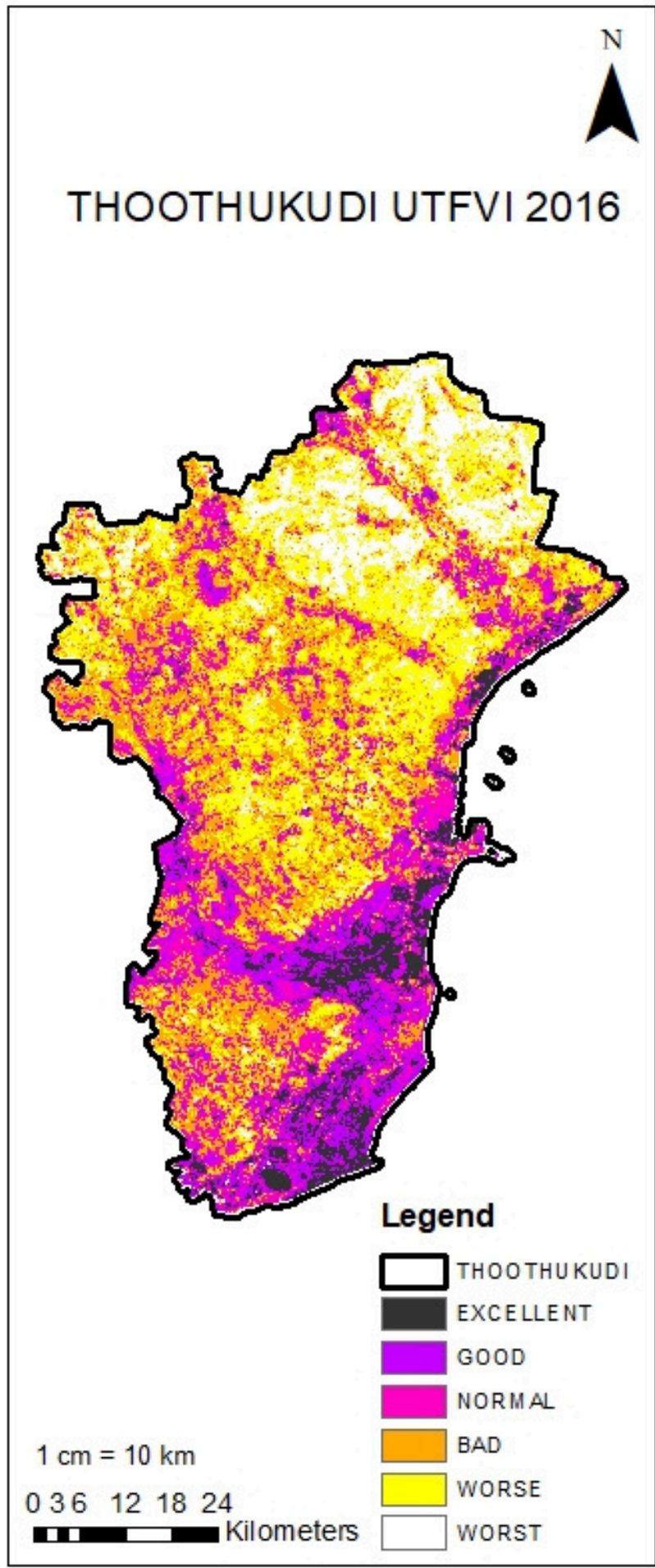
URBAN HEAT ISLAND



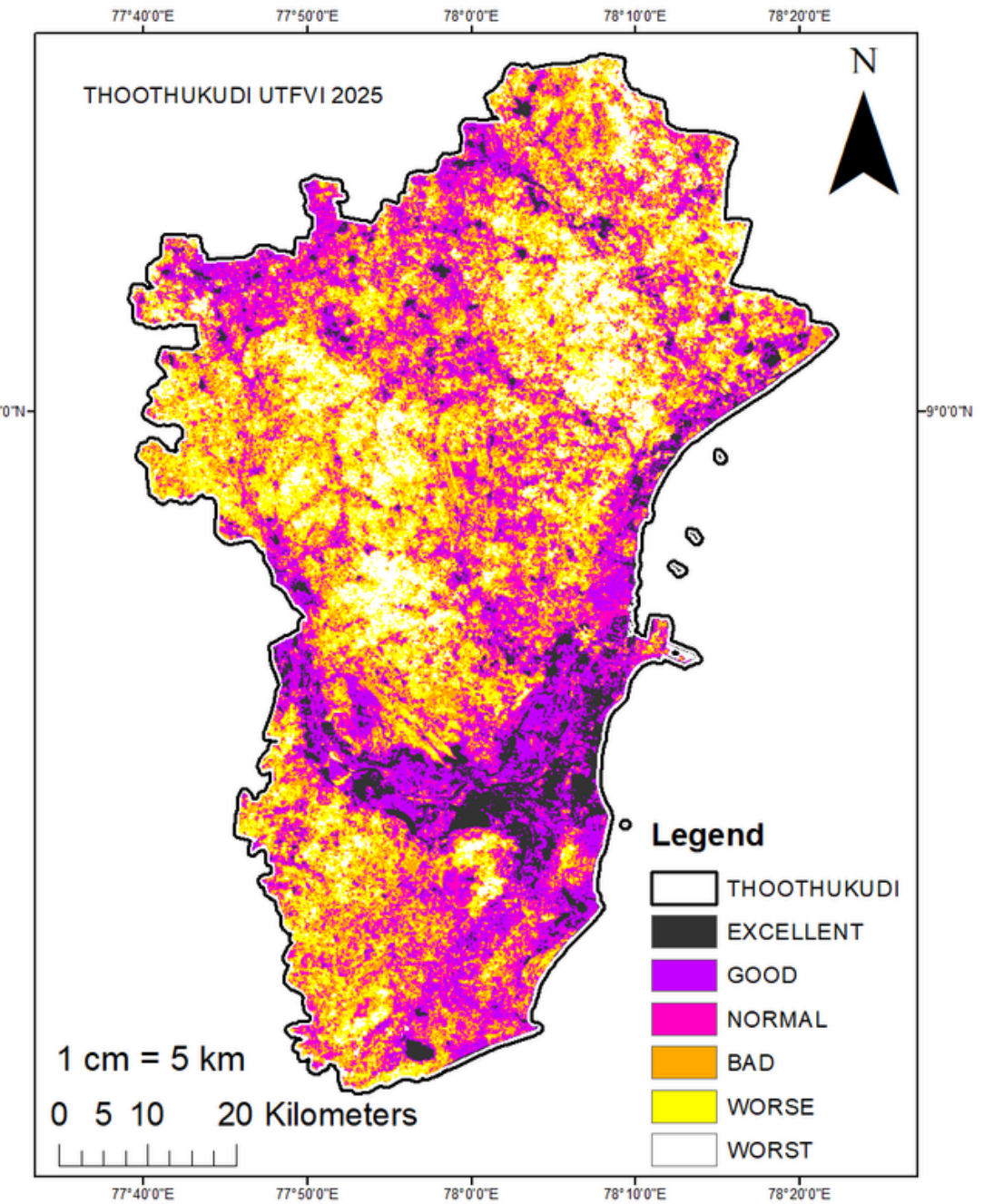
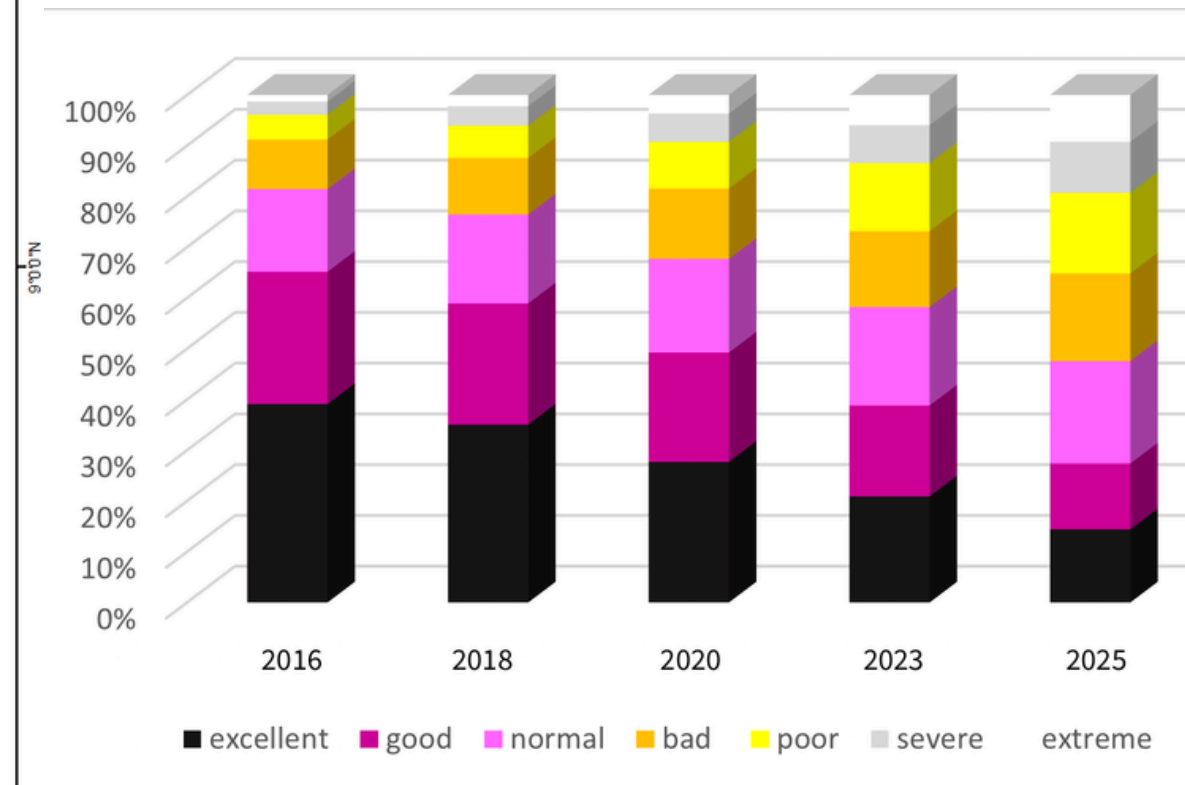
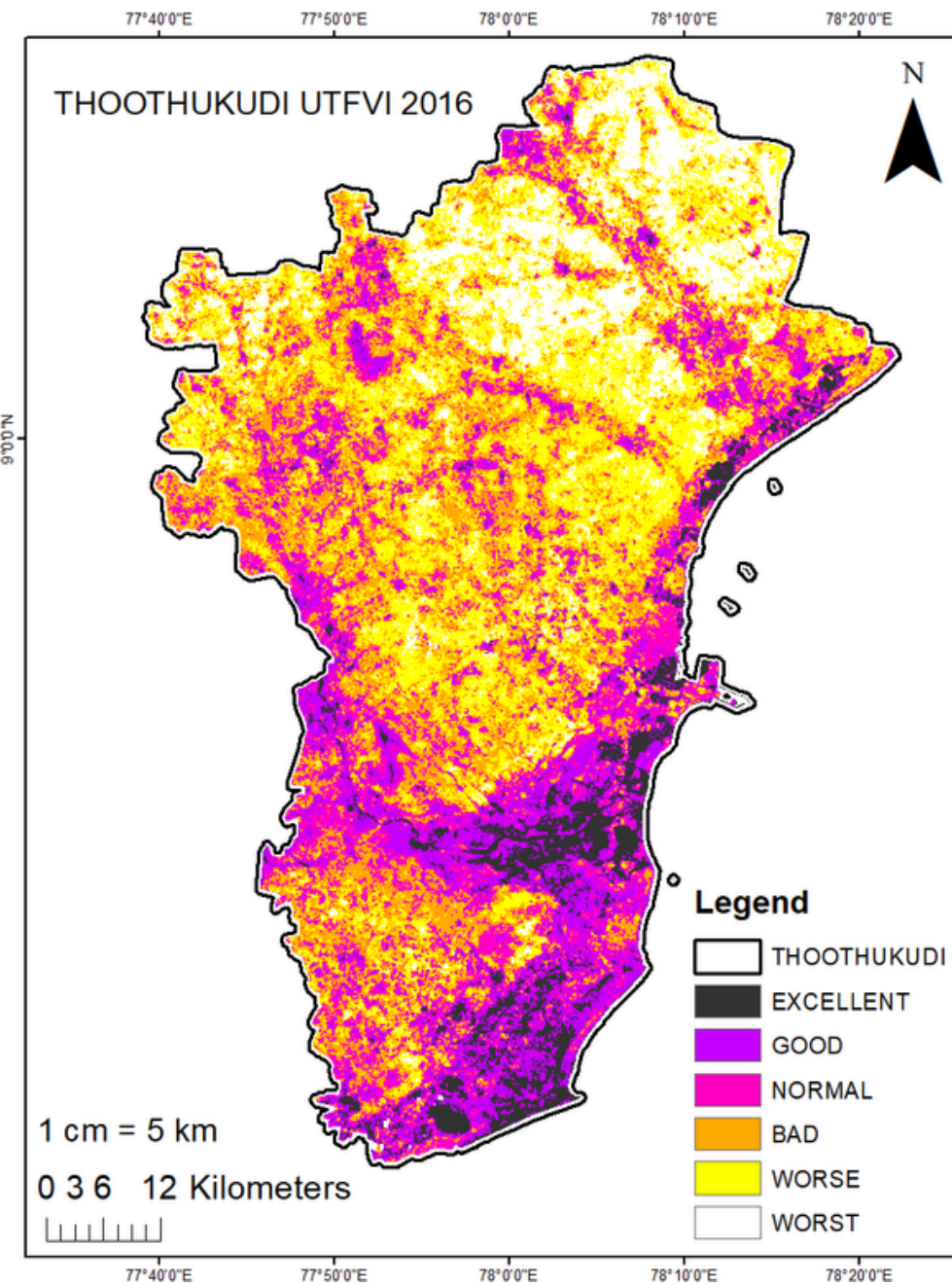
Urban Expansion (NDBI & UHI)

- NDBI:
 - 2016: Limited built-up areas.
 - 2021: Rapid urban expansion (higher NDBI values).
 - 2025: Saturation of built-up zones, with red dominating.
- UHI:
 - 2016: Mostly “cool” or “near mean.”
 - 2021: Warning zones appear.
 - 2025: Severe UHI dominates, showing urban heat stress due to concrete expansion and vegetation loss.

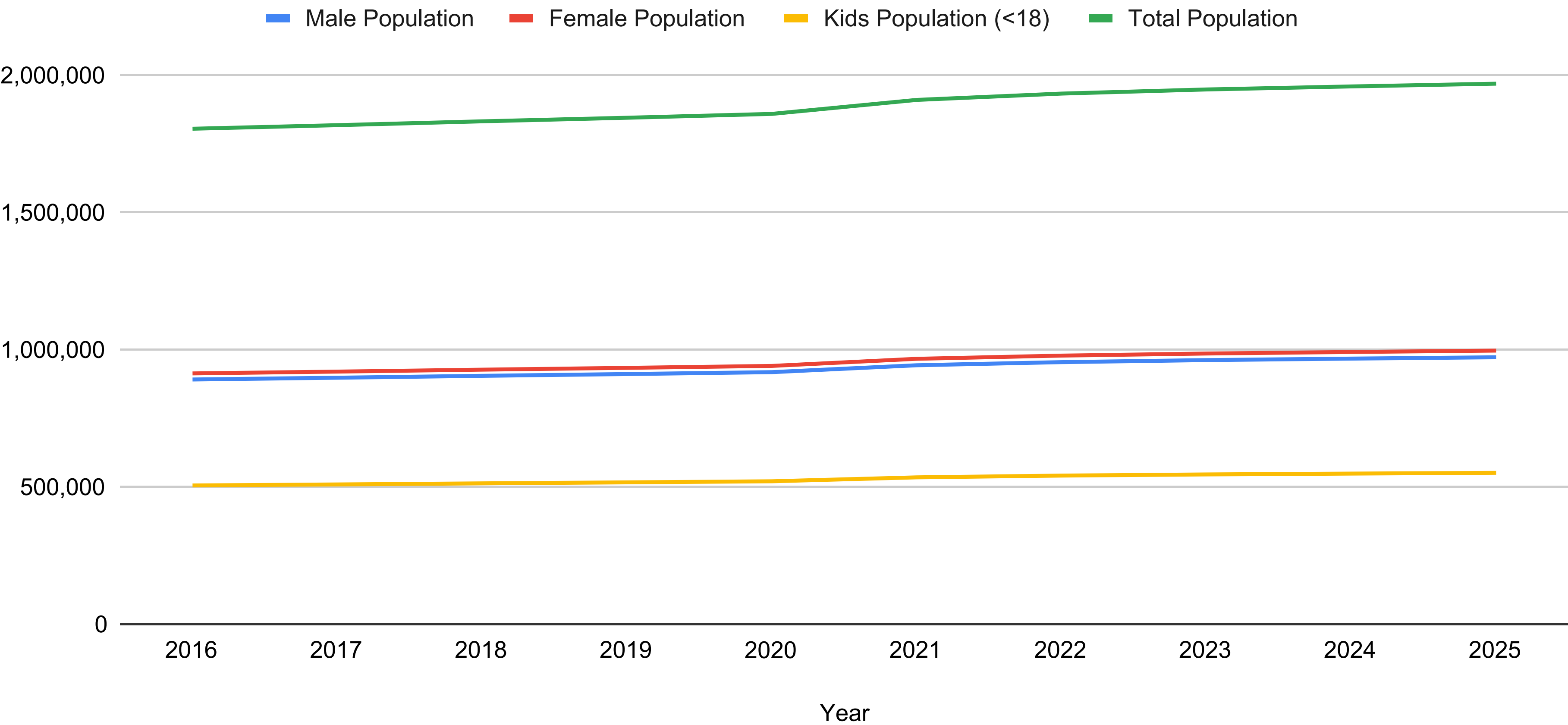
Implication: Built-up growth reduces natural cooling, amplifies heat stress, and worsens air quality. Human health risks (heat strokes, respiratory issues) rise sharply.



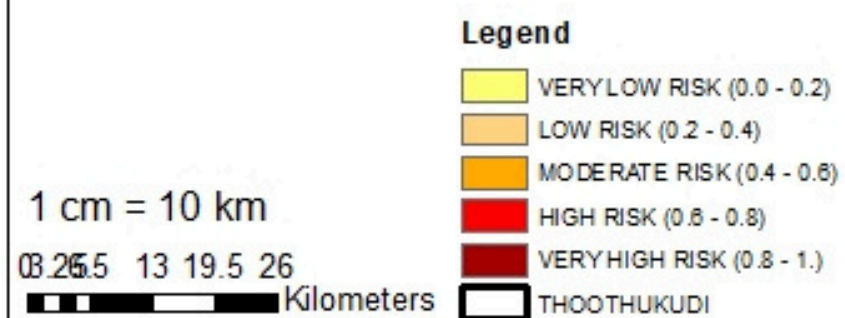
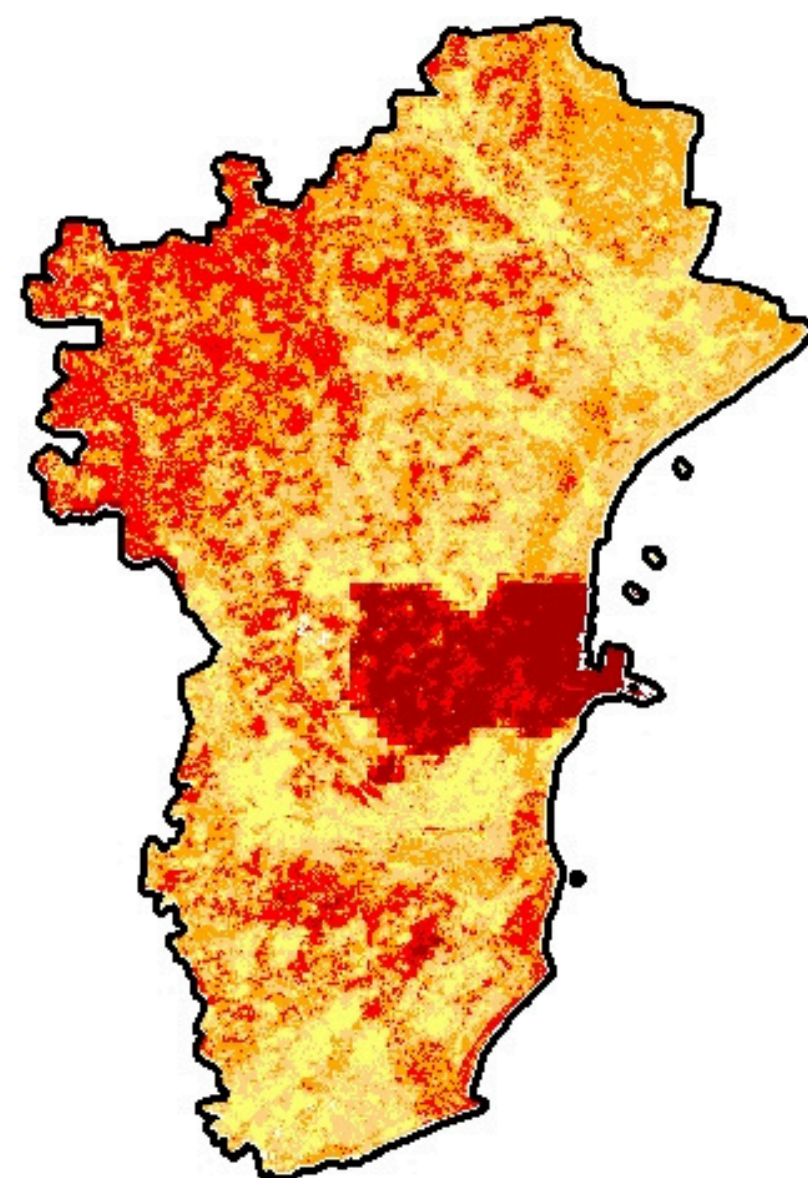
URBAN THERMAL FIELD VARIANCE INDEX



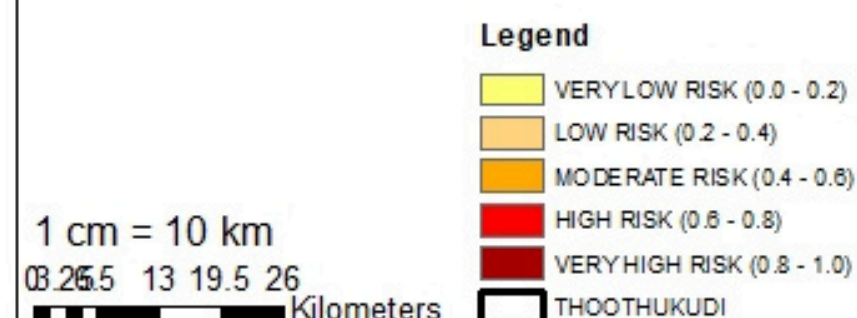
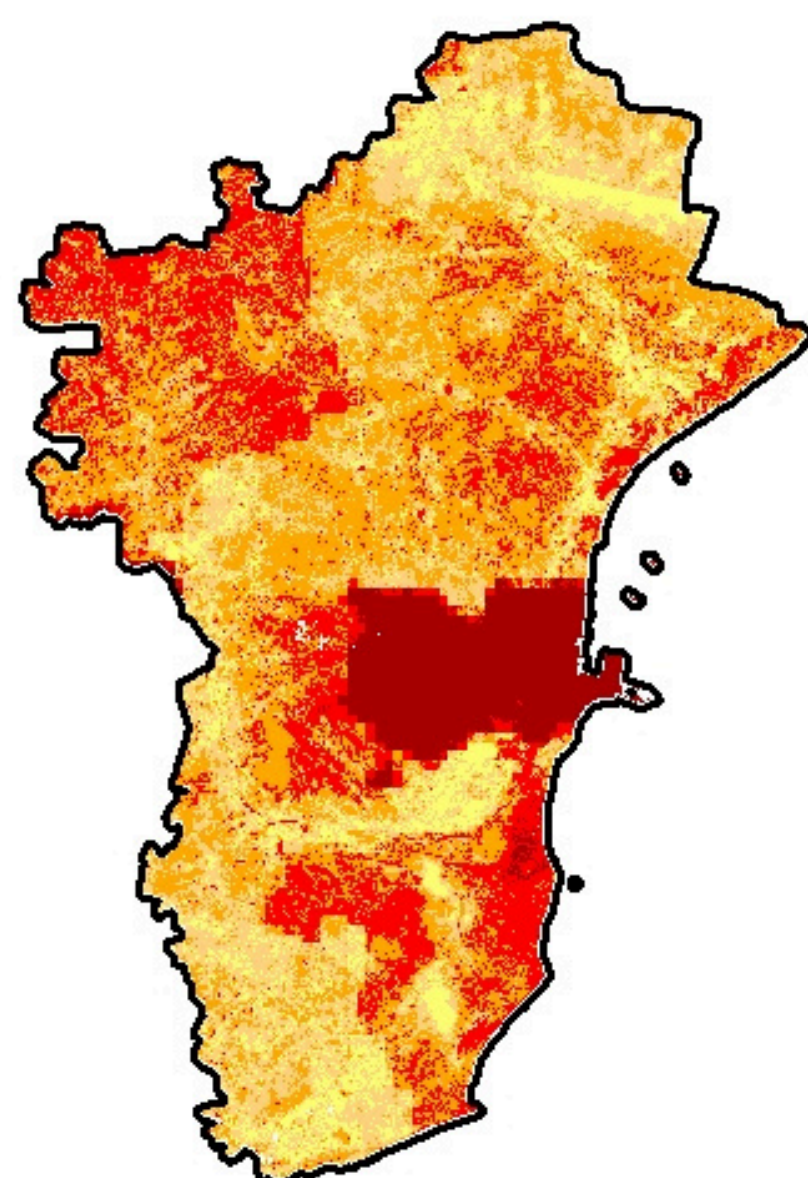
Male Population, Female Population, Kids Population (<18) and Total Population



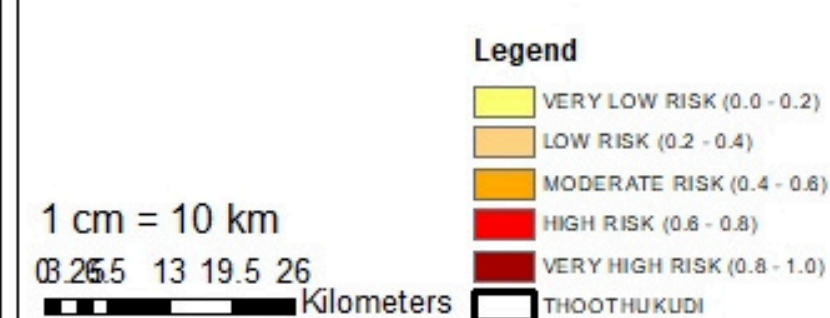
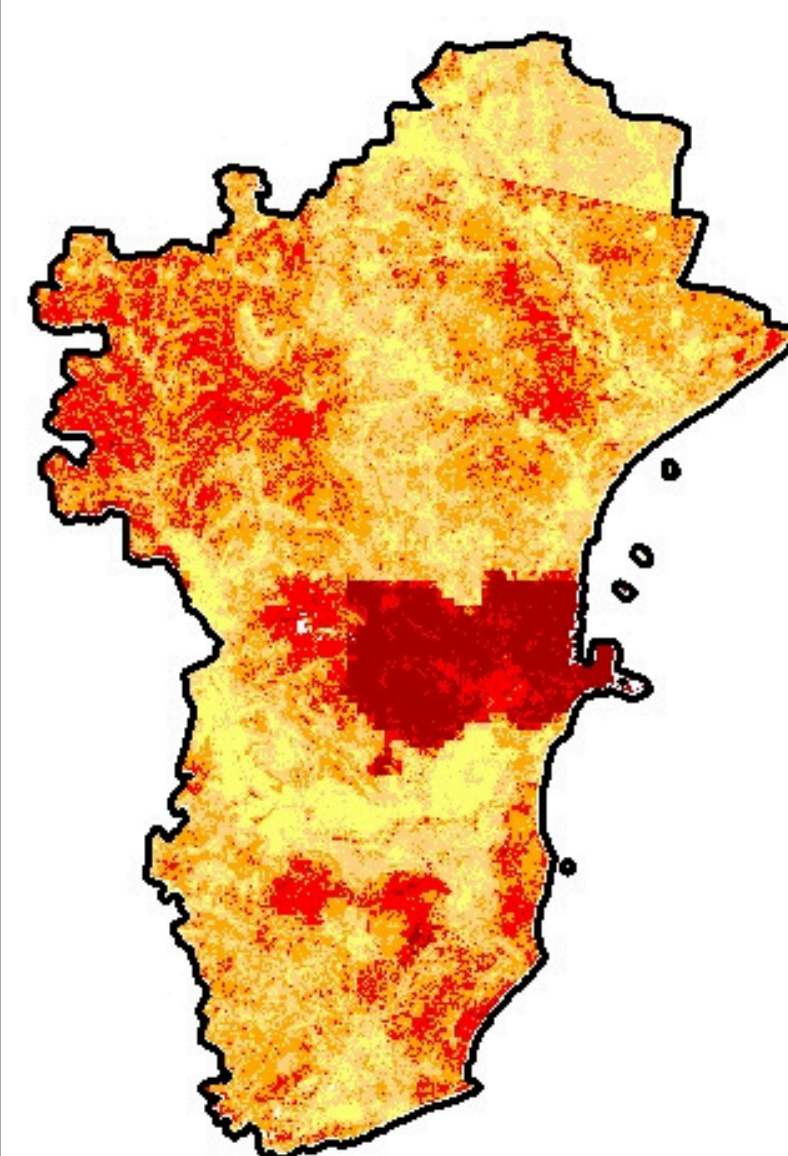
THOOTHUKUDI HHRI 2016



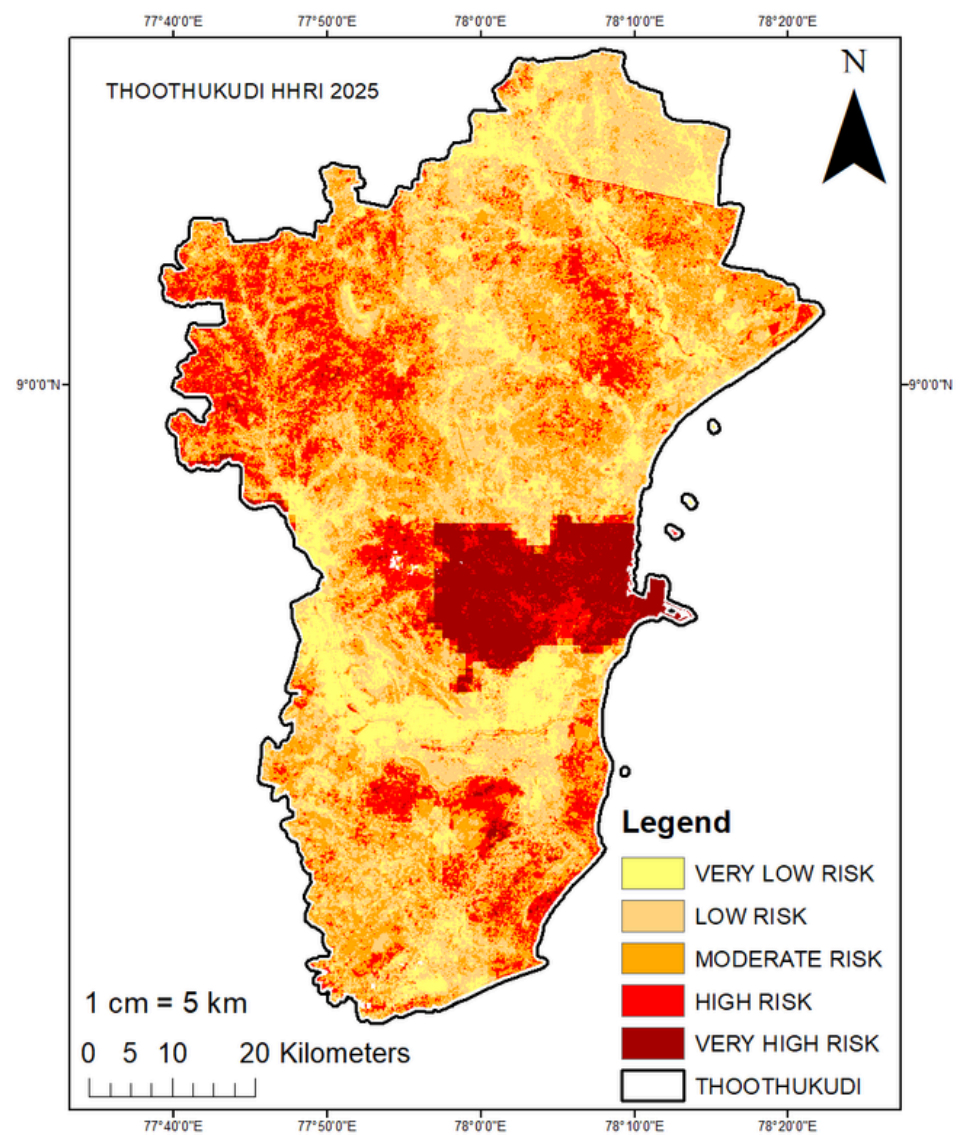
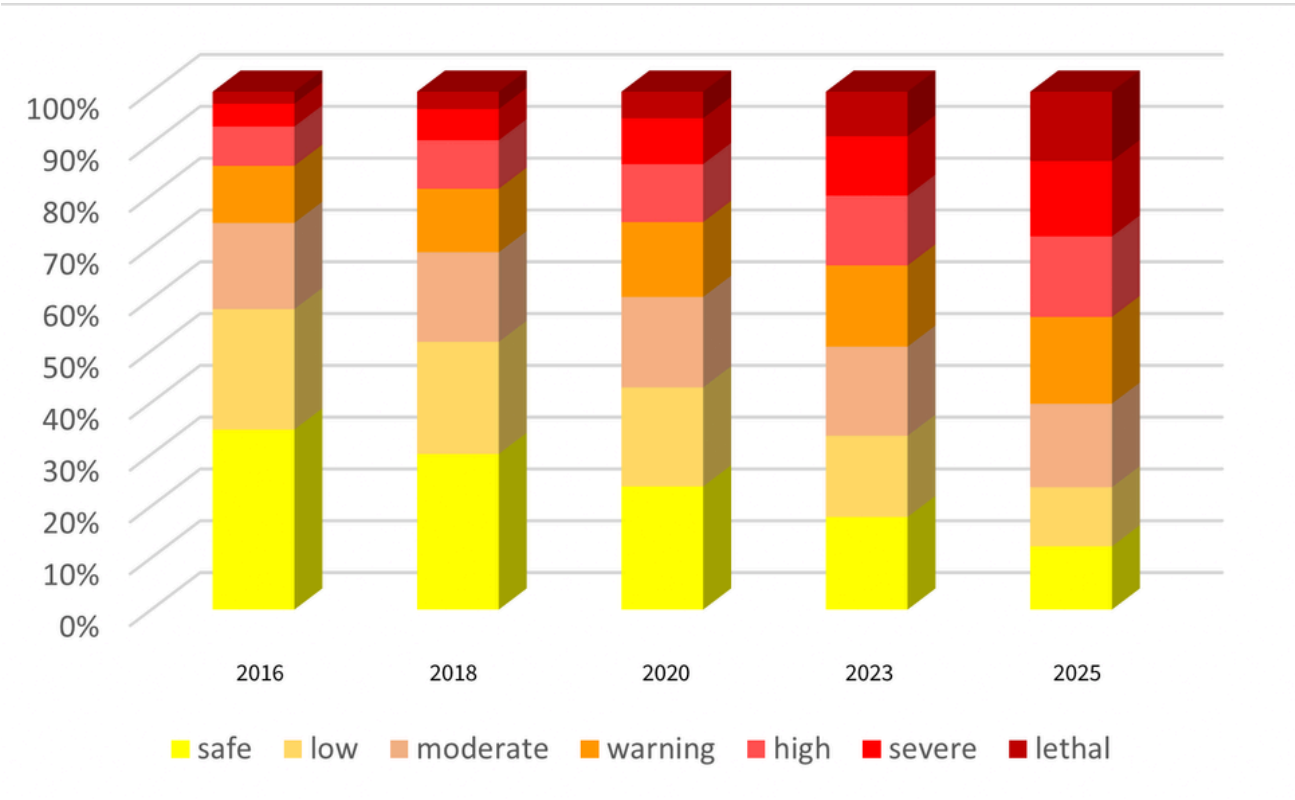
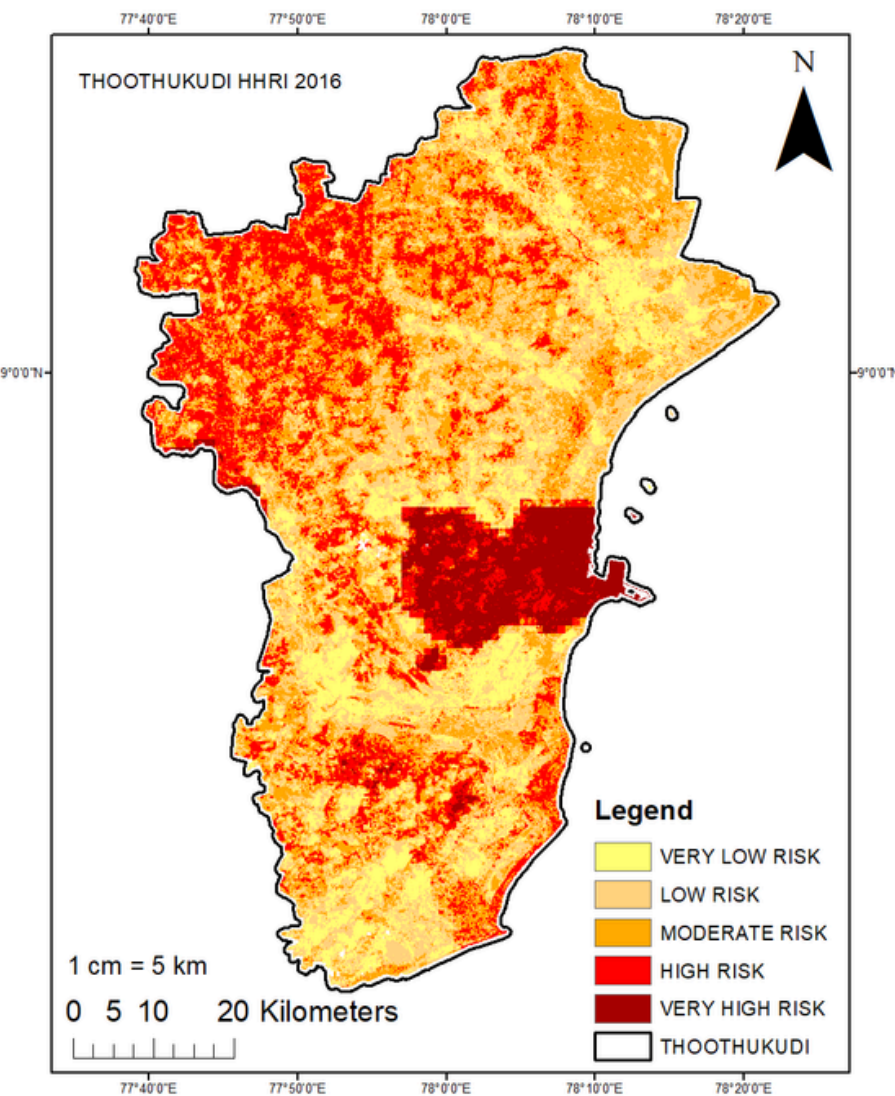
THOOTHUKUDI HHRI 2021



THOOTHUKUDI HHRI 2025



HEAT HEALTH RISK INDEX



RESULT

our maps clearly show a progressive ecological deterioration:

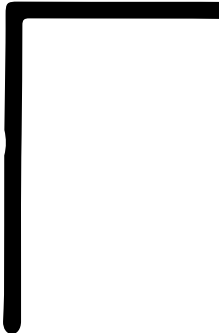
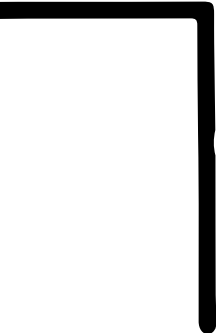
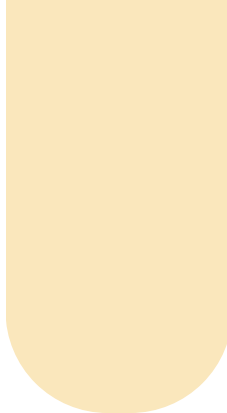
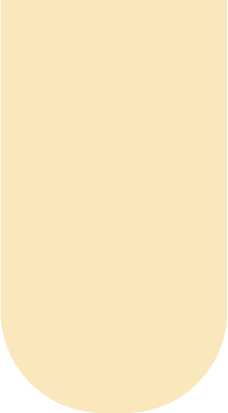
- 2016: Manageable stress, balanced vegetation.
- 2021: Transition phase — urban growth + rising stress.
- 2025: Critical stage — ecological collapse with severe

heat, vegetation loss, and human health risks.

This analysis can be framed in our project as a cause-effect chain: Temperature rise → Vegetation stress → Urban heat amplification → Human health risk → Ecological worst condition.

RESULT

In conclusion, Thoothukudi has experienced a marked **increase in built-up area** and a corresponding **decline in vegetation** and **open spaces** over the past decade. This land transformation has directly intensified land surface temperature and strengthened urban heat island effects. If these trends continue, the region may **face severe environmental and socio-economic challenges**. Therefore, immediate implementation of sustainable **urban planning** and **climate-resilient strategies** is essential to ensure long-term ecological balance and urban sustainability.



THANK YOU